Lab 5 Solution

Build, Test, and Package Applications with IntelliJ IDEA and Docker

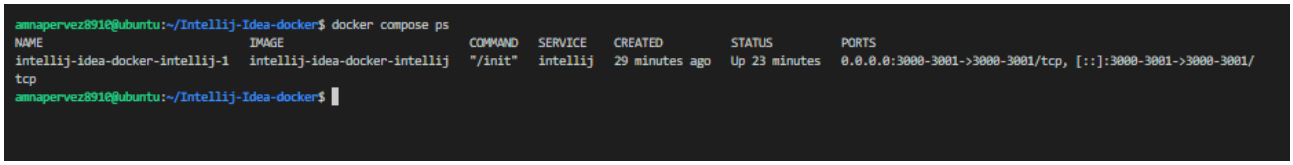
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Registration No.	2023-BSCS-10
Course	CS-649: Cloud DevOps
Section	A
GitHub Username	amnapervez8910
Submission Directory	DevOps/Labs/Lab05

Screenshots for Lab 5.

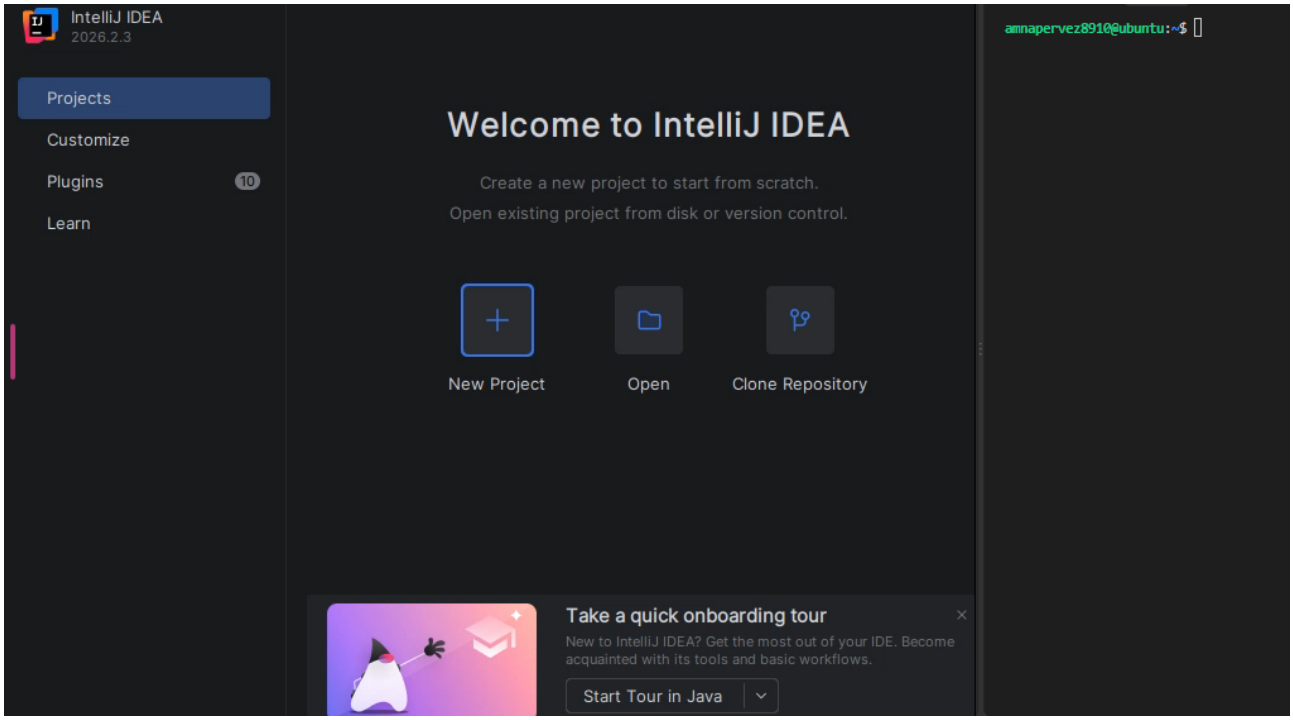
Task 1: Clone and Launch IntelliJ IDEA

```
amnapervez8910@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Intellij-Idea-docker.git
Cloning into 'Intellij-Idea-docker'...
remote: Enumerating objects: 110, done.
remote: Counting objects: 100% (110/110), done.
remote: Compressing objects: 100% (96/96), done.
remote: Total 110 (delta 54), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (110/110), 43.18 KiB | 7.20 MiB/s, done.
Resolving deltas: 100% (54/54), done.
amnapervez8910@ubuntu:~$
```

intellij_repo_clone.png
Cloned the Intellij-Idea-docker repository using git clone on my Ubuntu terminal.

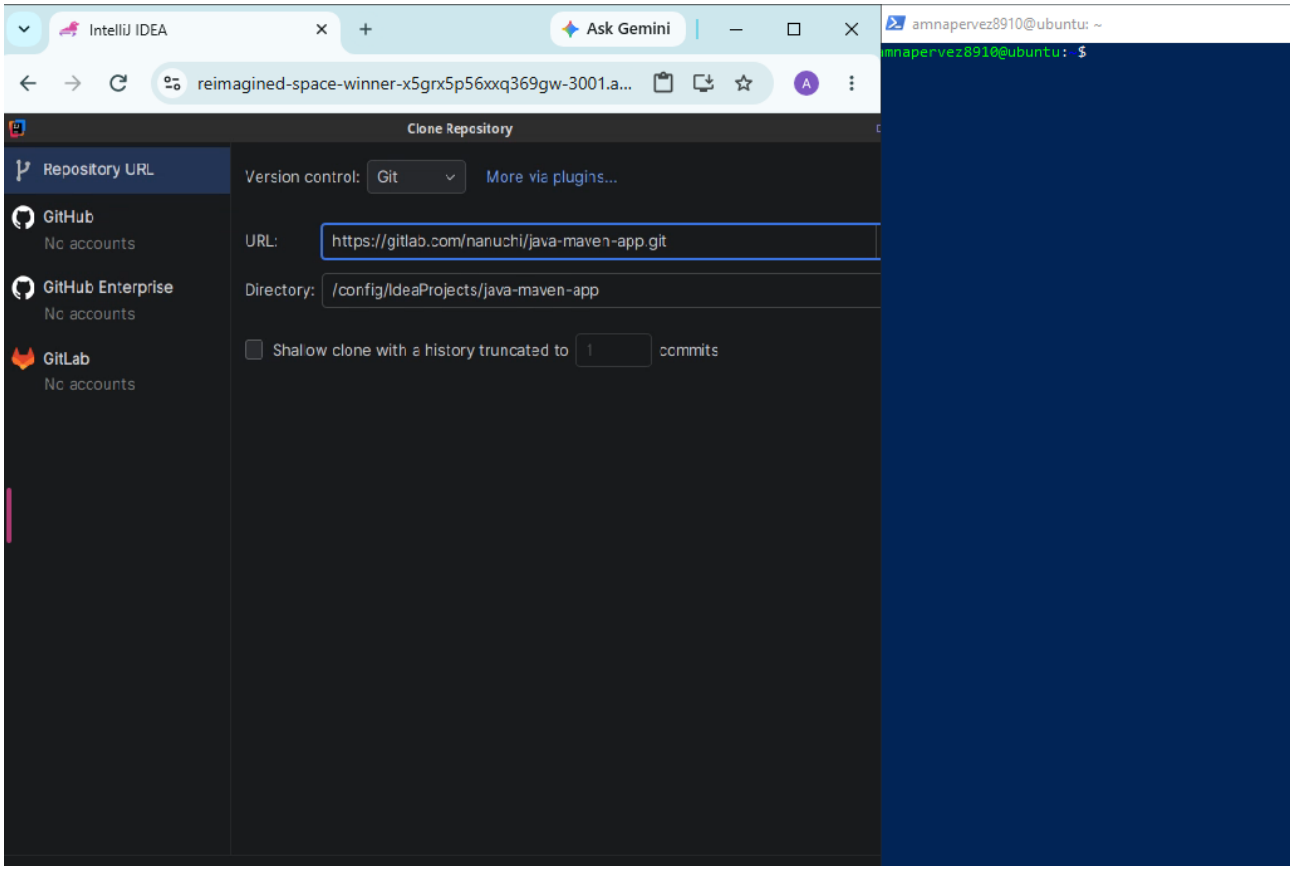


intellij_running.png
docker compose ps shows the intellij service is running on port 3001.

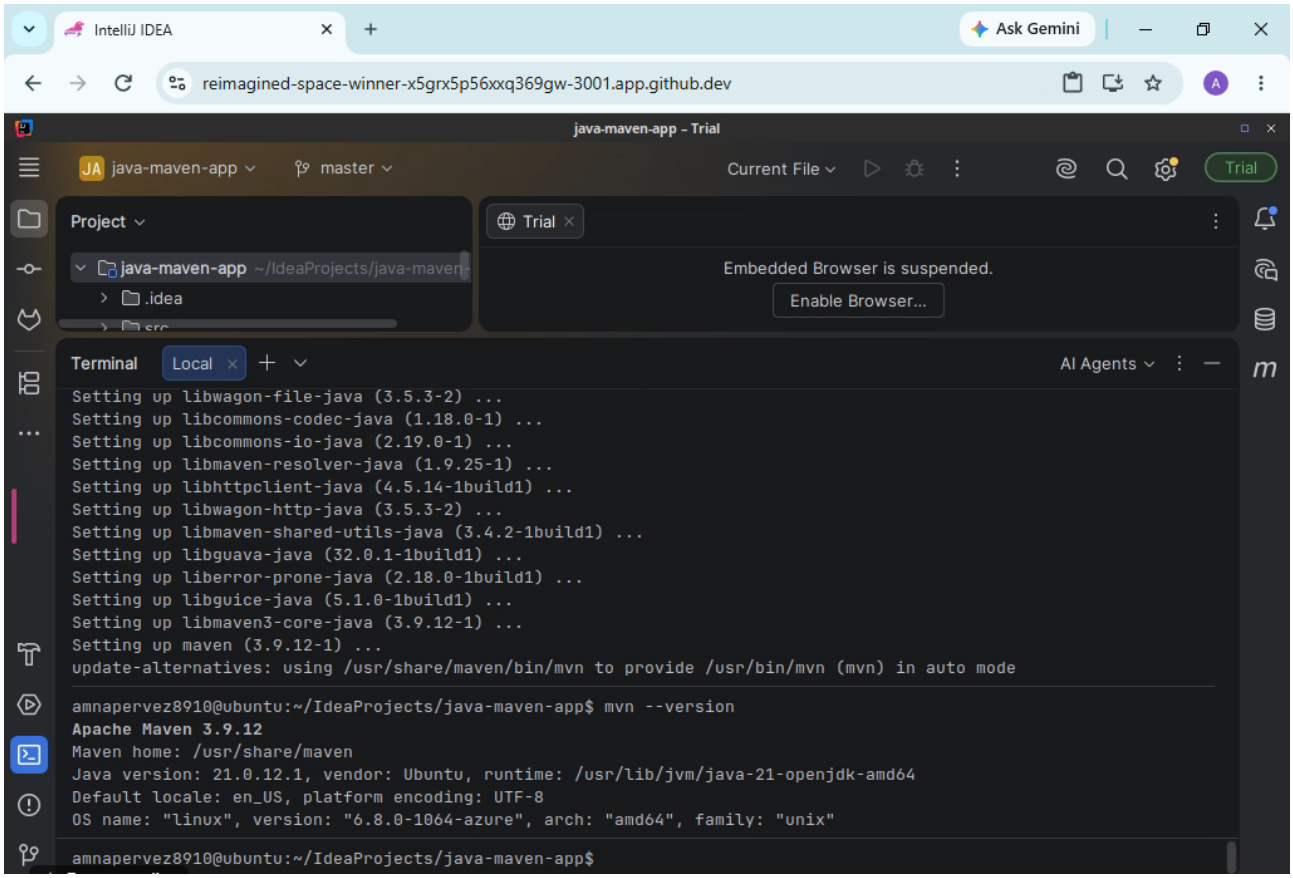


intellij_desktop.png
IntelliJ IDEA welcome screen open in the browser desktop, with my Ubuntu terminal beside it.

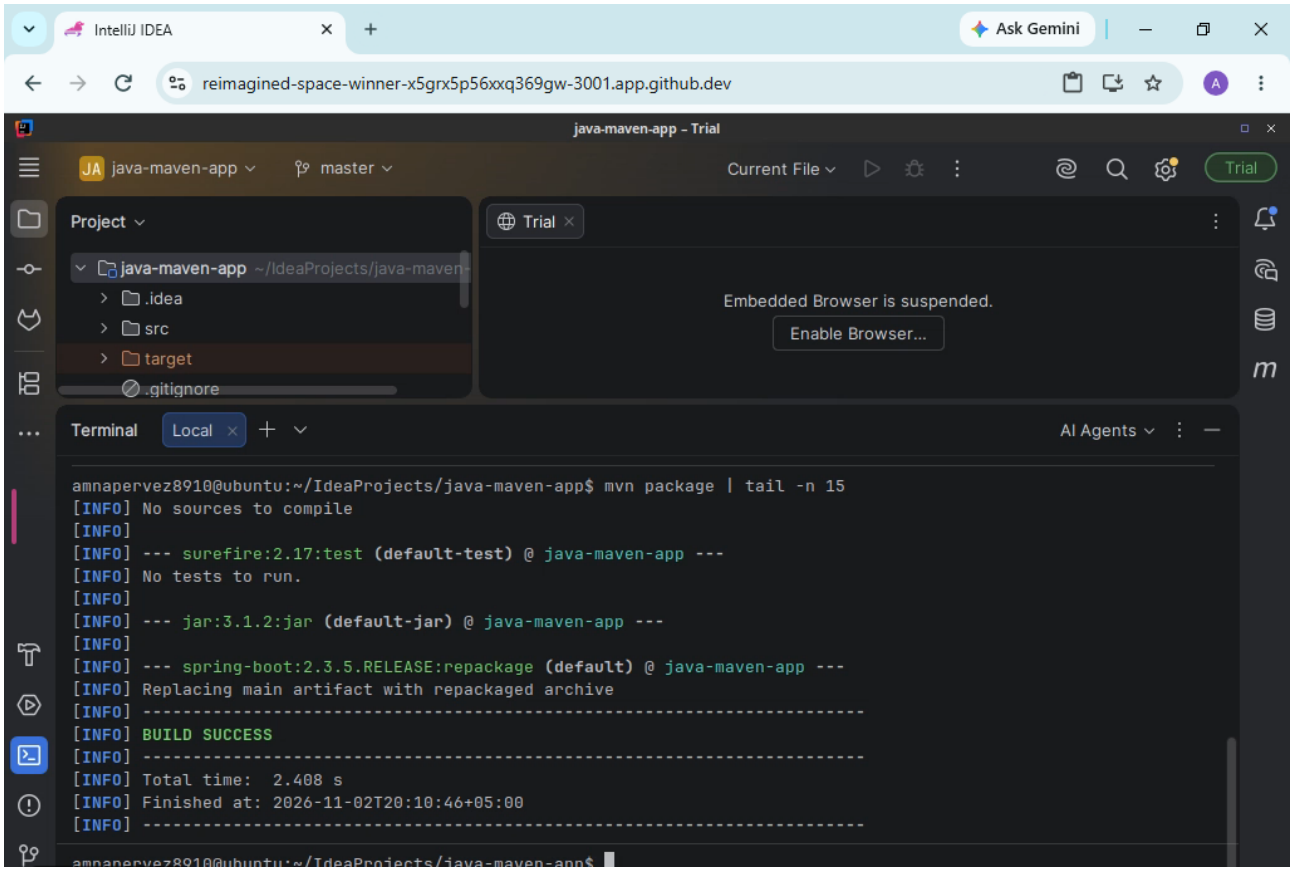
Task 2: Build and Run the Maven Project



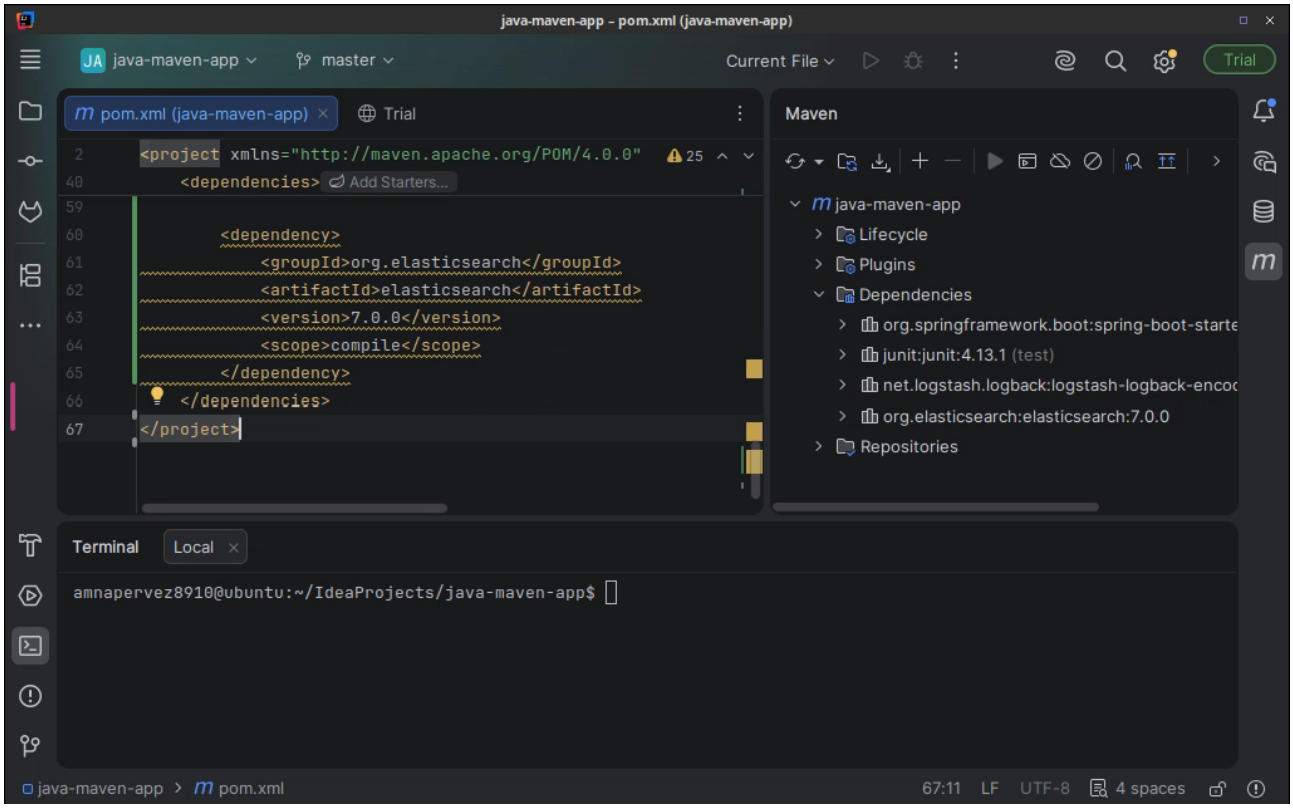
maven_project_clone.png
Cloning the java-maven-app project from GitLab into /config/IdeaProjects/java-maven-app.



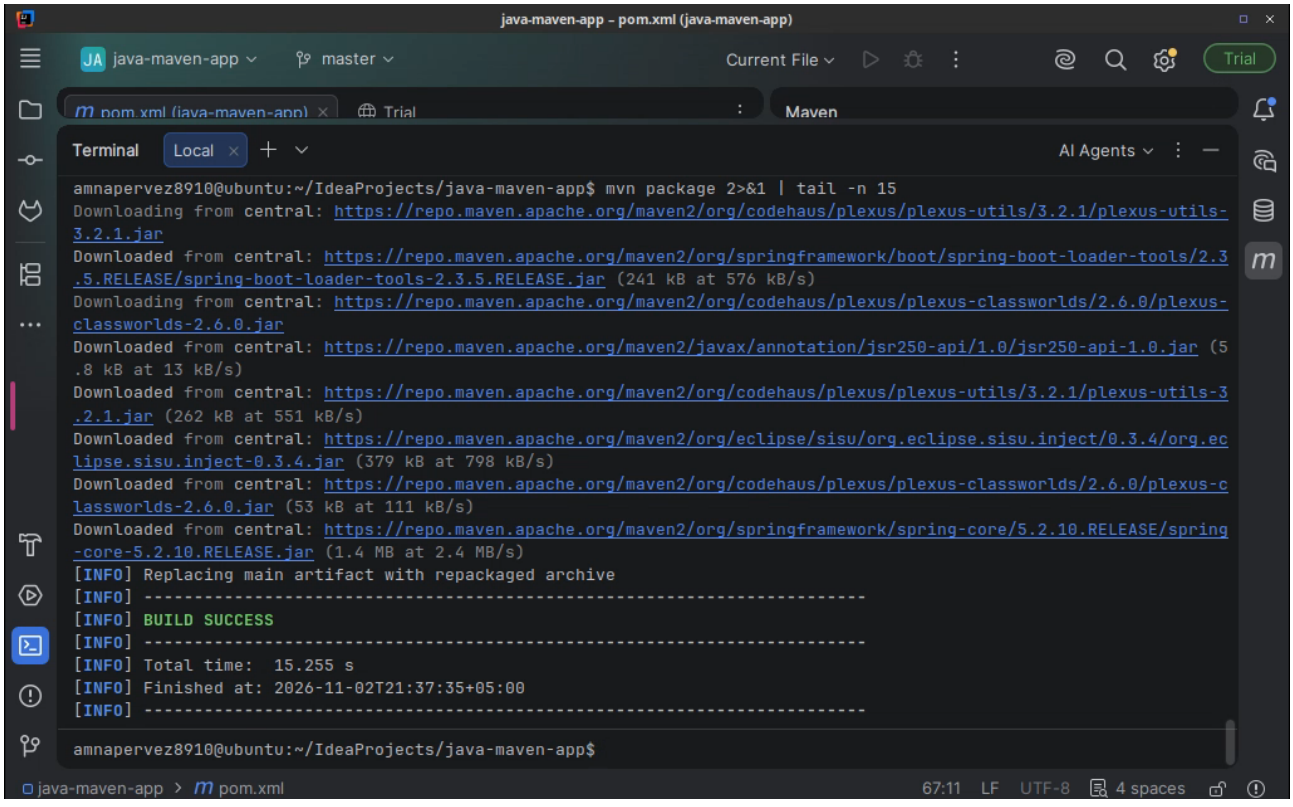
maven_installed.png
Installed Maven and checked the version with mvn --version (Apache Maven 3.9.12).



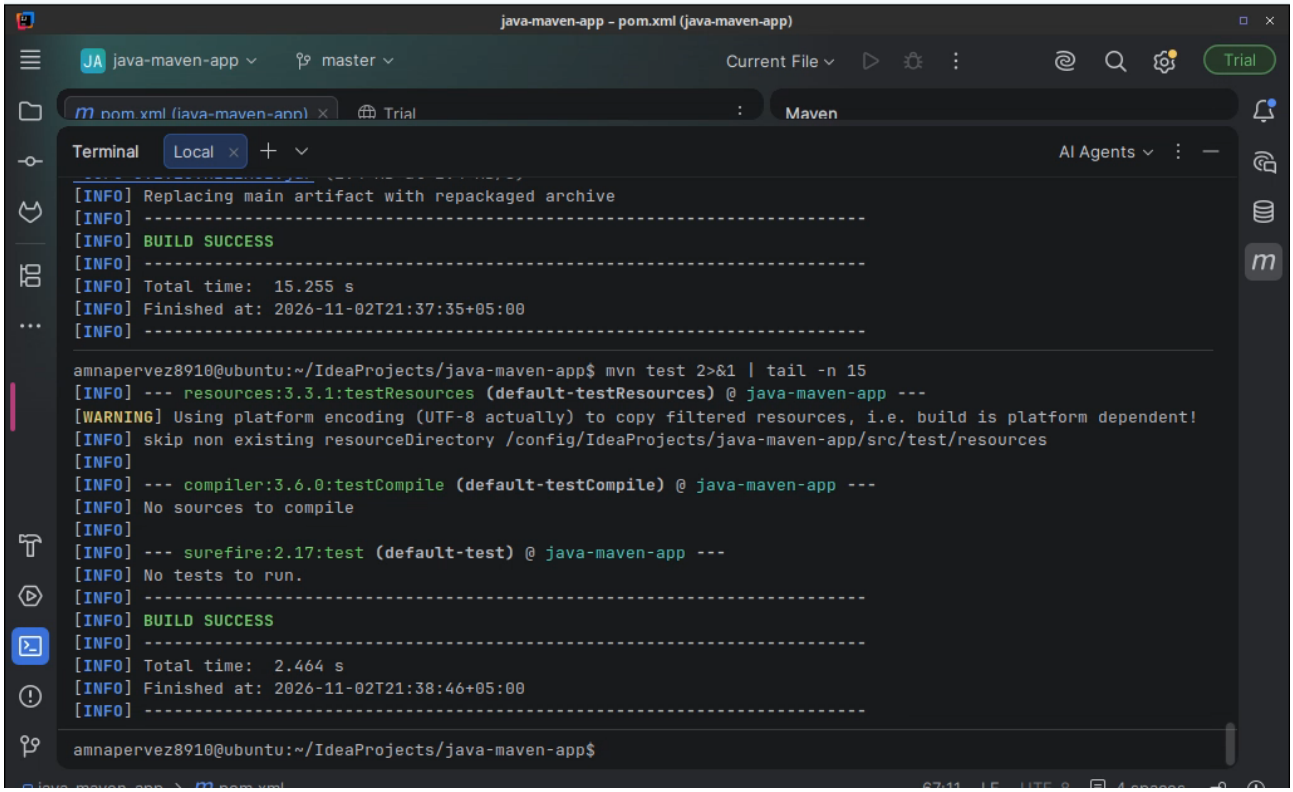
maven_package.png
Ran mvn package and got BUILD SUCCESS.



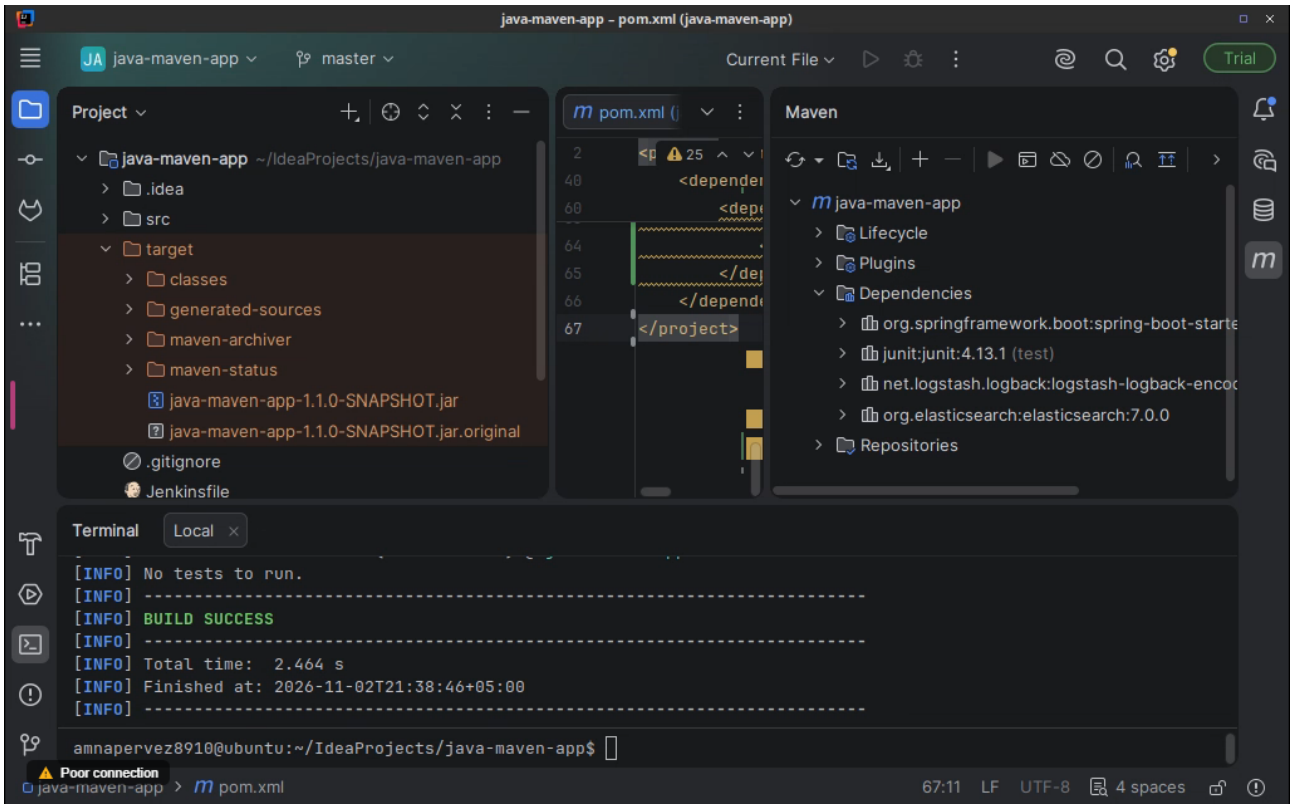
maven_elasticsearch_dependency.png
Added the Elasticsearch 7.0.0 dependency in pom.xml and synced Maven. It shows up in the Maven Dependencies list.



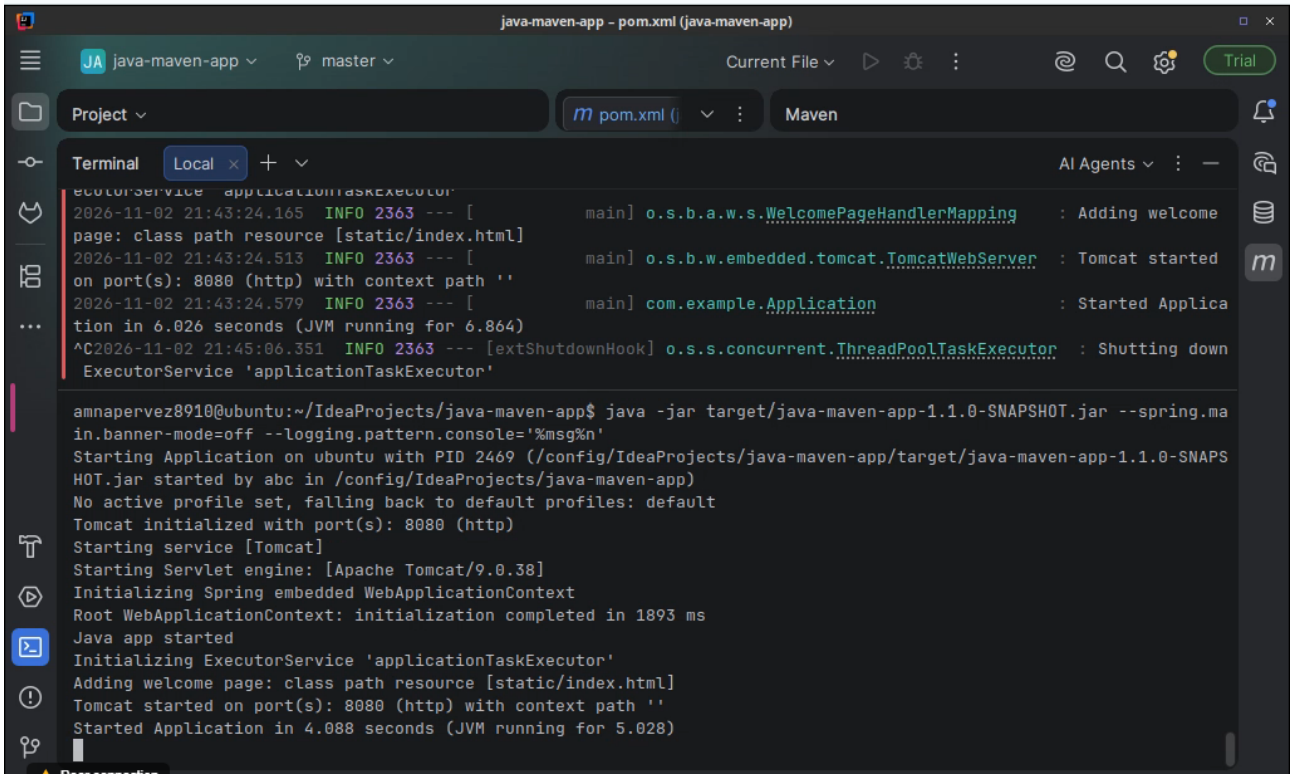
maven_repackage.png
Ran mvn package again after adding the dependency. BUILD SUCCESS again.



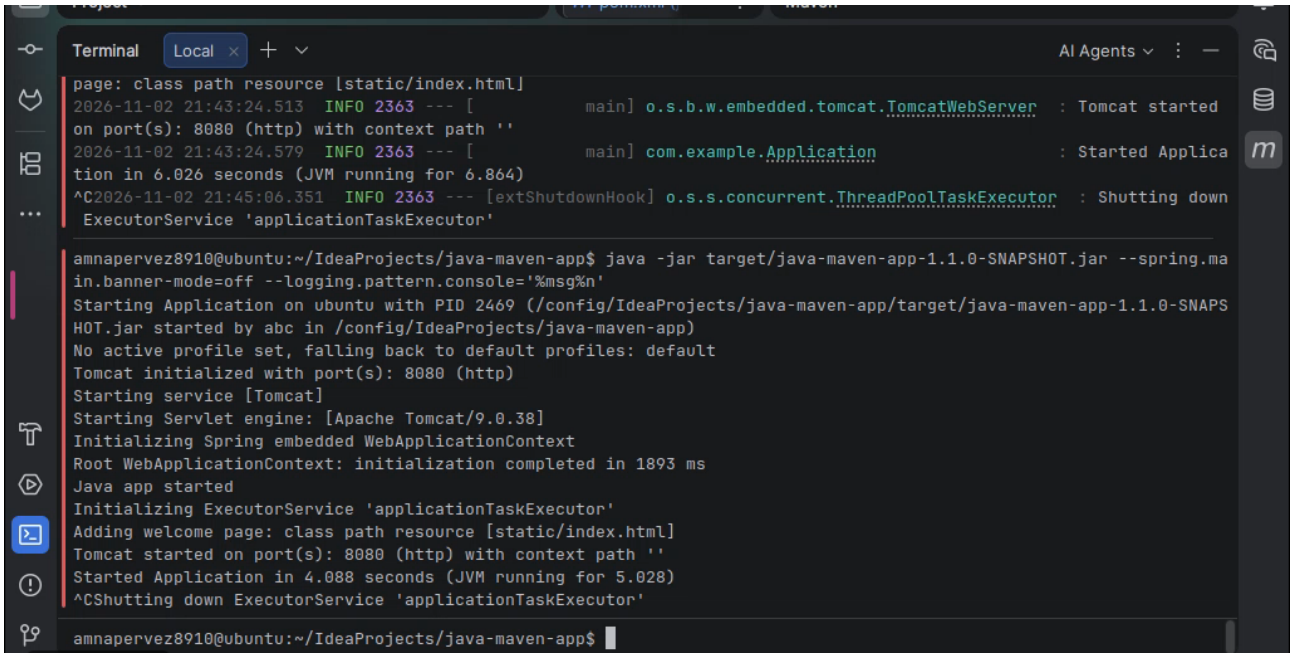
maven_test.png
Ran mvn test. It says No tests to run because the project has no tests, and the build is successful.



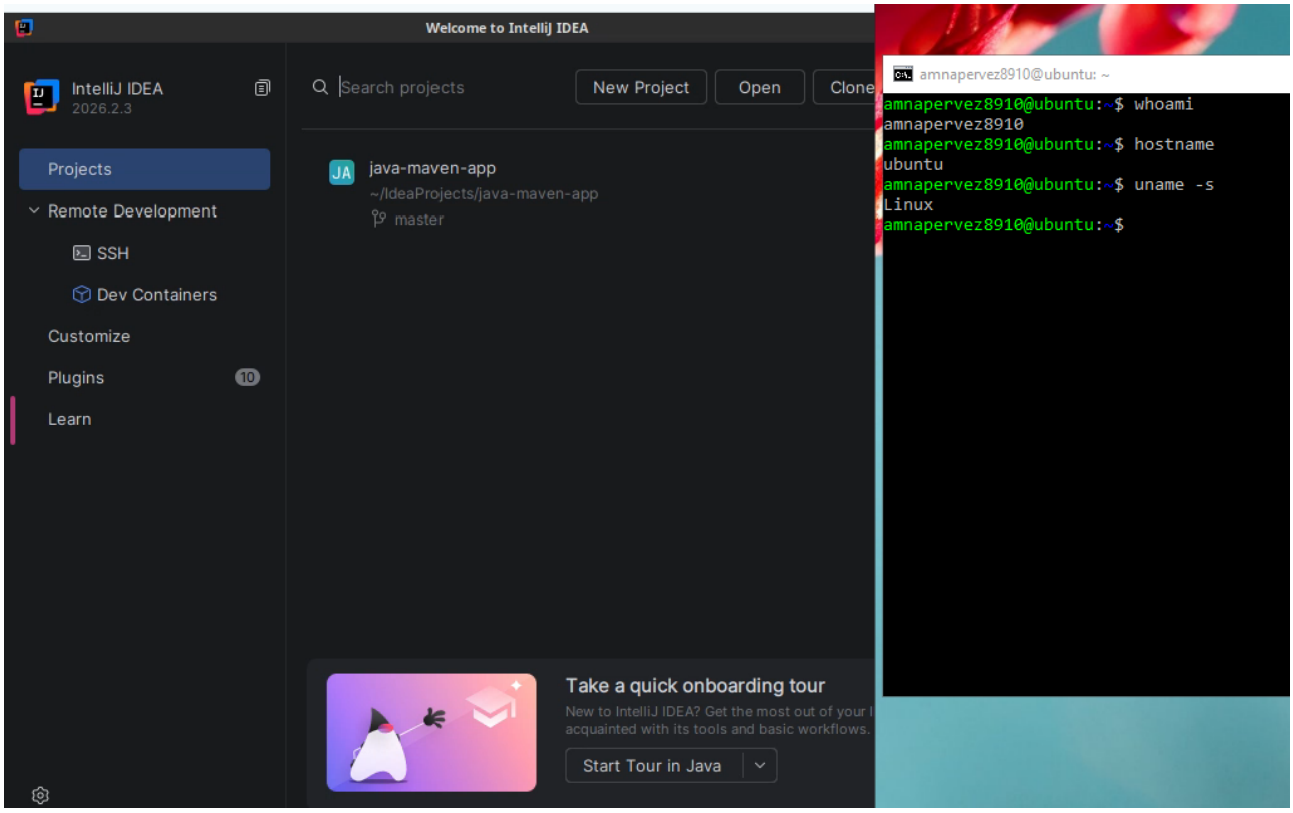
maven_jar.png
The target folder has java-maven-app-1.1.0-SNAPSHOT.jar.



maven_running.png
Ran the jar with java -jar. Tomcat started on port 8080.

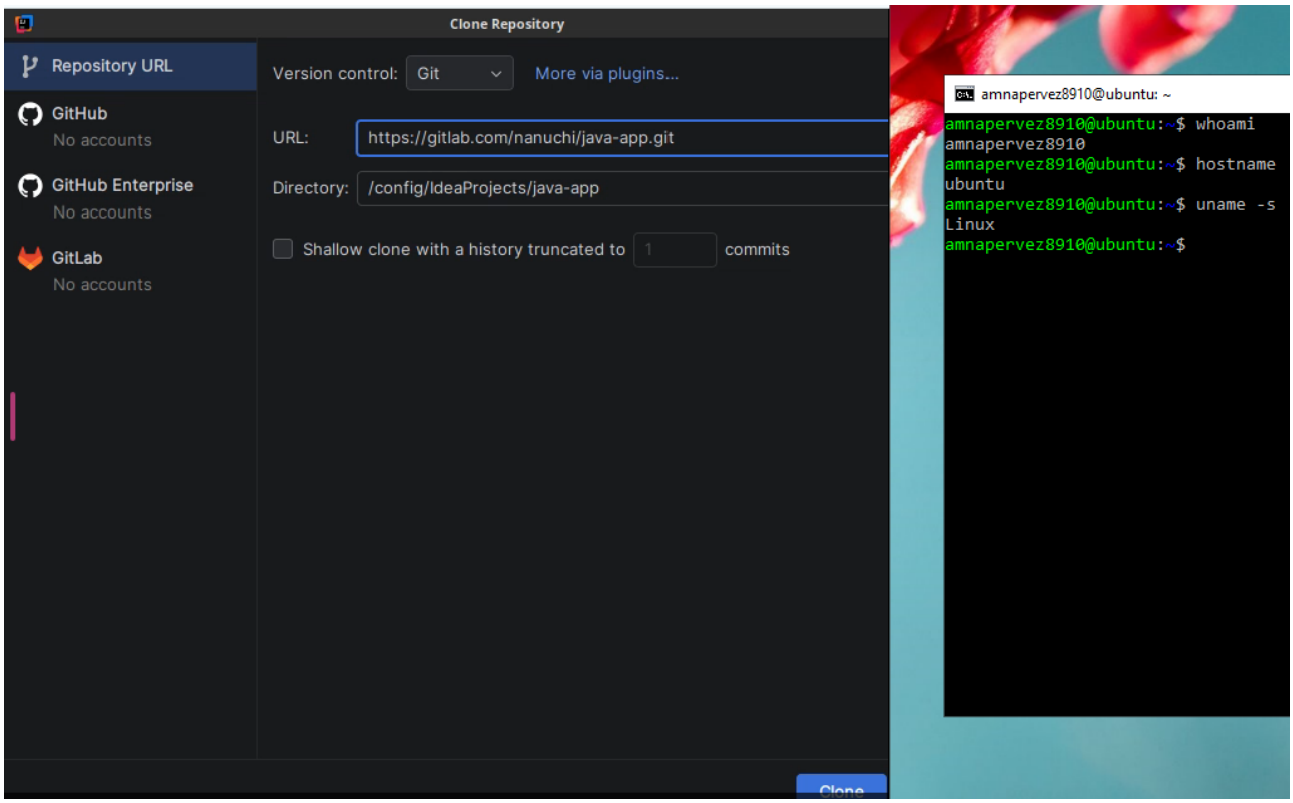


maven_stopped.png
Pressed Ctrl+C to stop the application. The prompt came back.

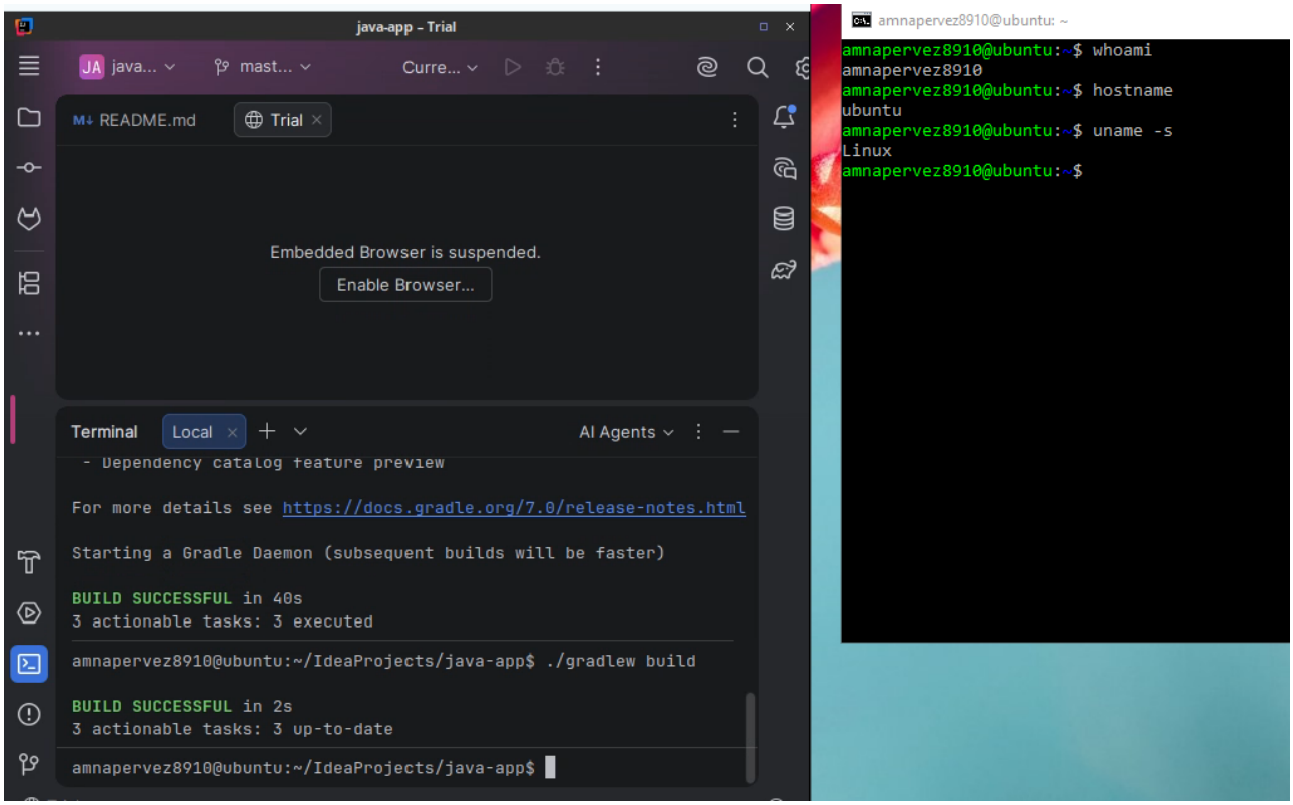


maven_project_closed.png
Closed the project with File > Close Project and returned to the Welcome screen.

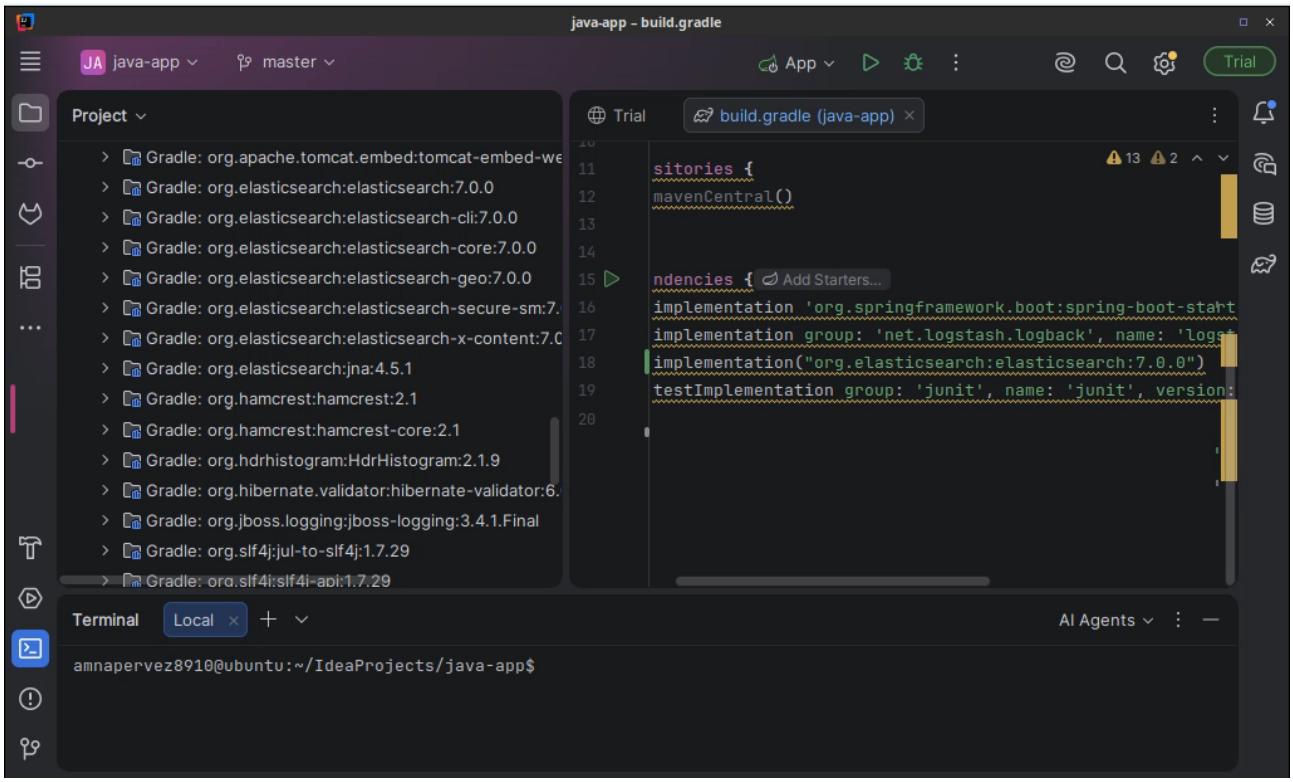
Task 3: Build and Run the Gradle Project



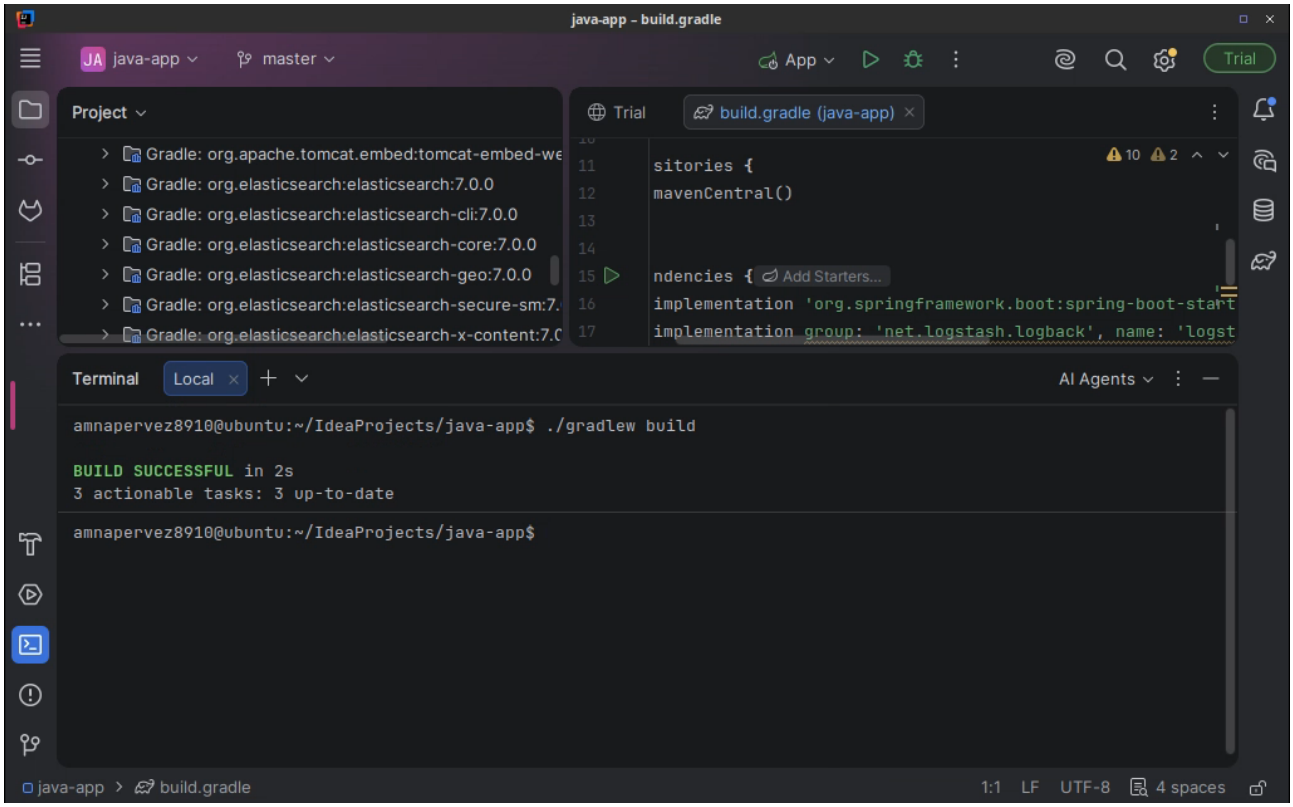
gradle_project_clone.png
Cloning the java-app project from GitLab into /config/IdeaProjects/java-app.



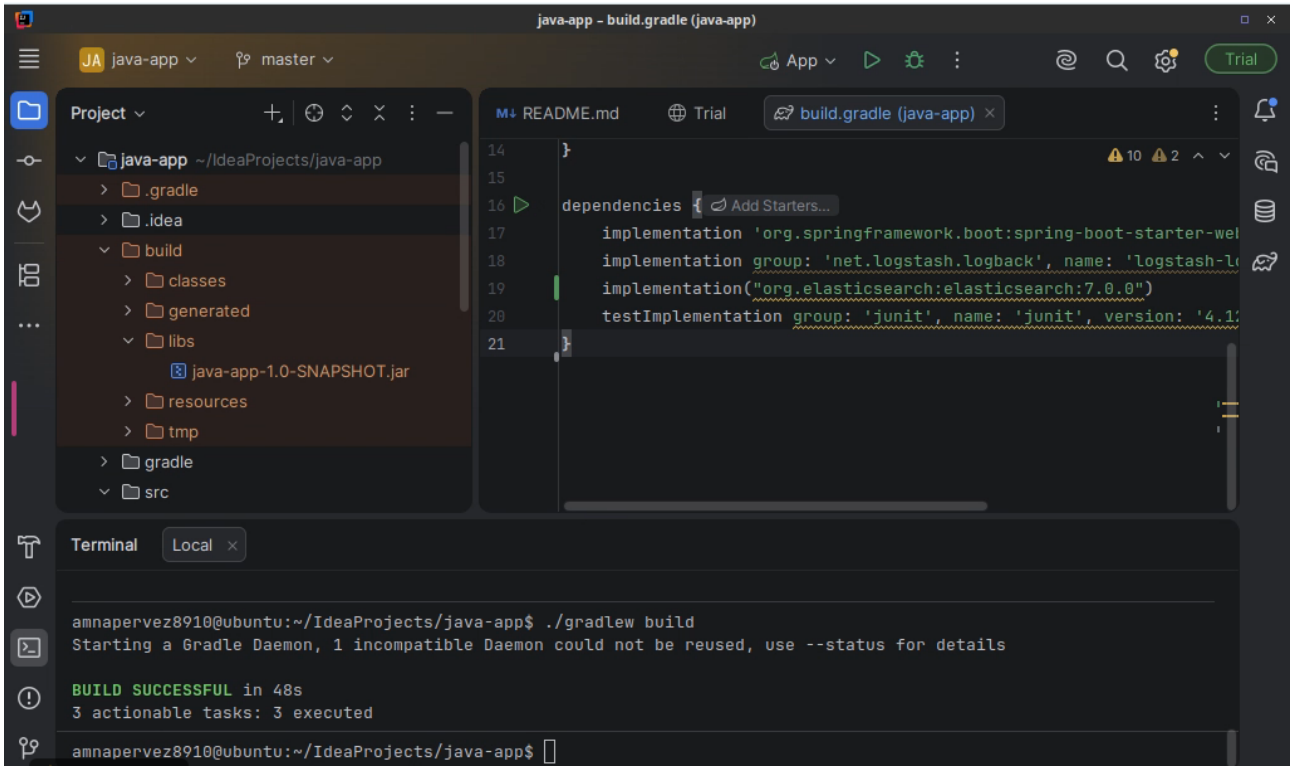
gradle_build.png
Ran ./gradlew build. BUILD SUCCESSFUL.



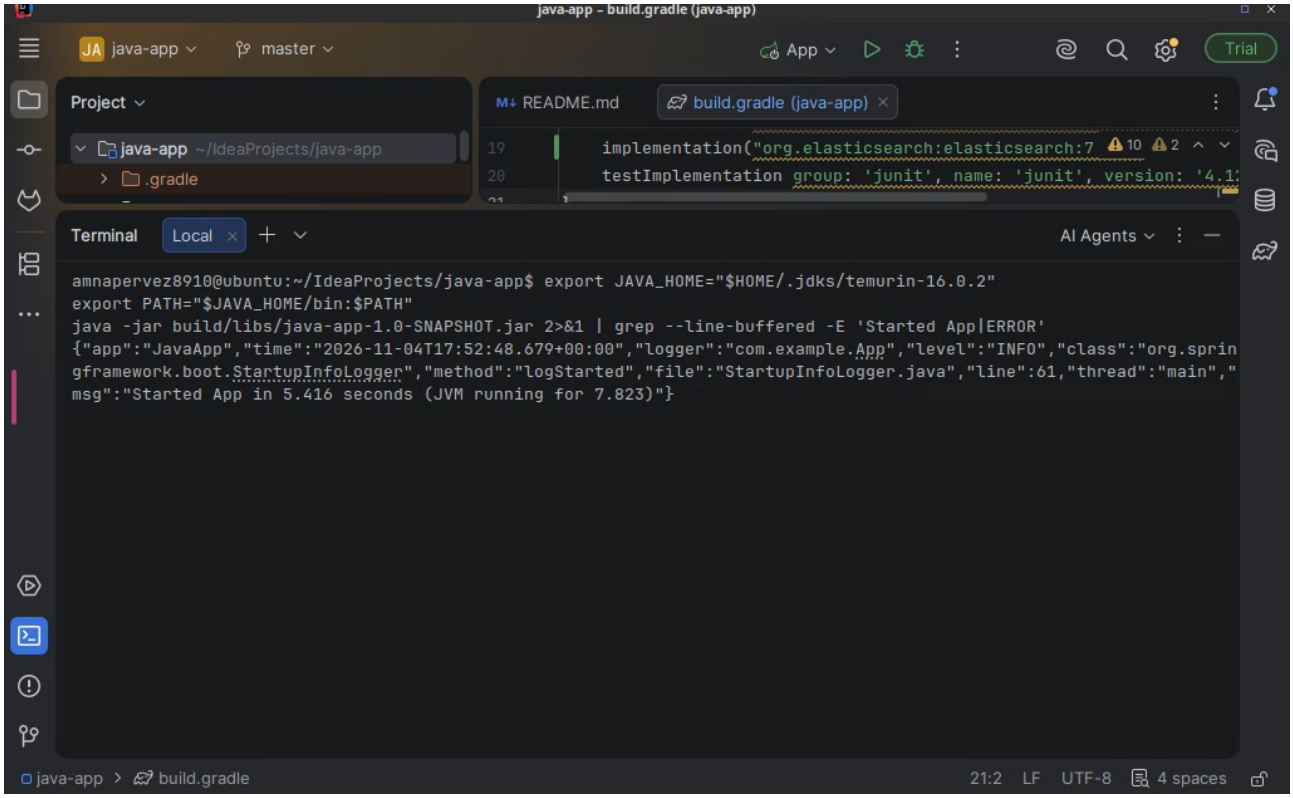
gradle_elasticsearch_dependency.png
Added the Elasticsearch 7.0.0 line in build.gradle and synced. The library is listed under External Libraries.



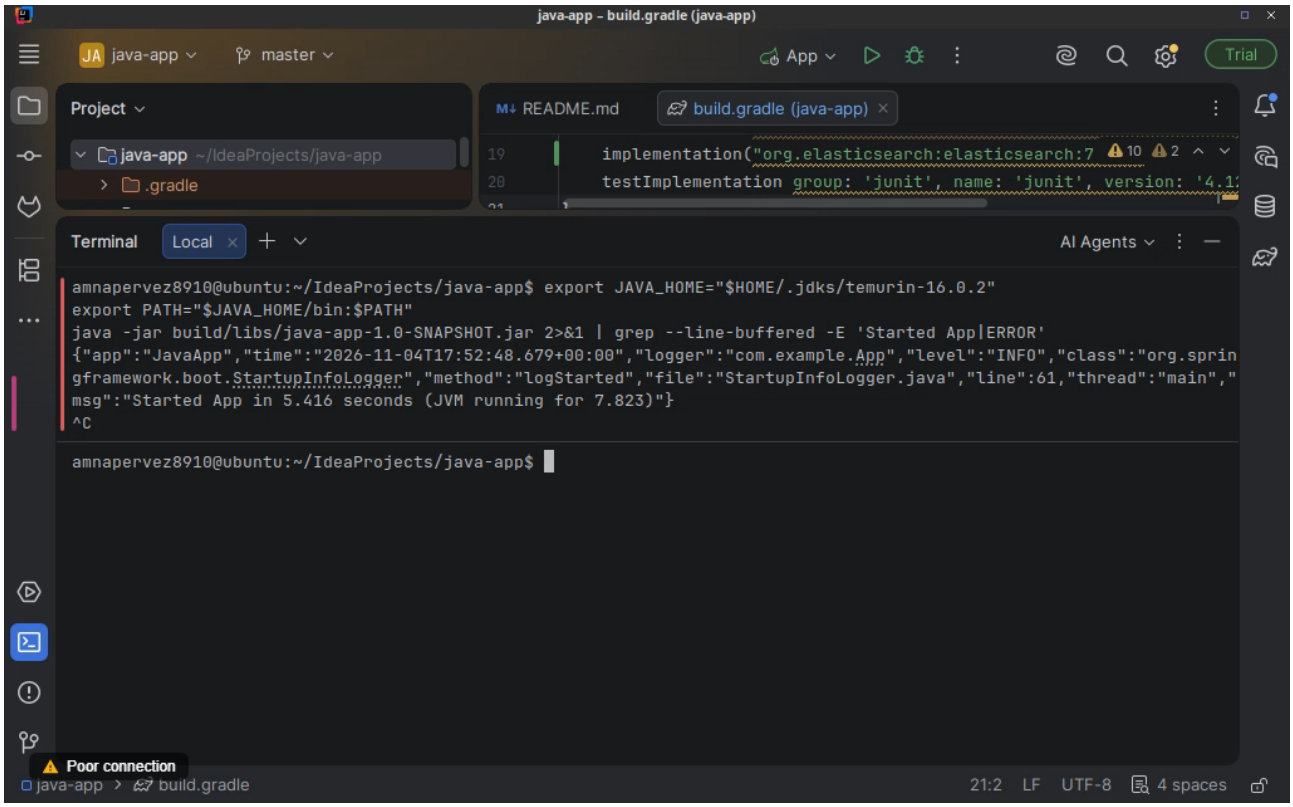
gradle_rebuild.png
Ran ./gradlew build again after adding the dependency. BUILD SUCCESSFUL.



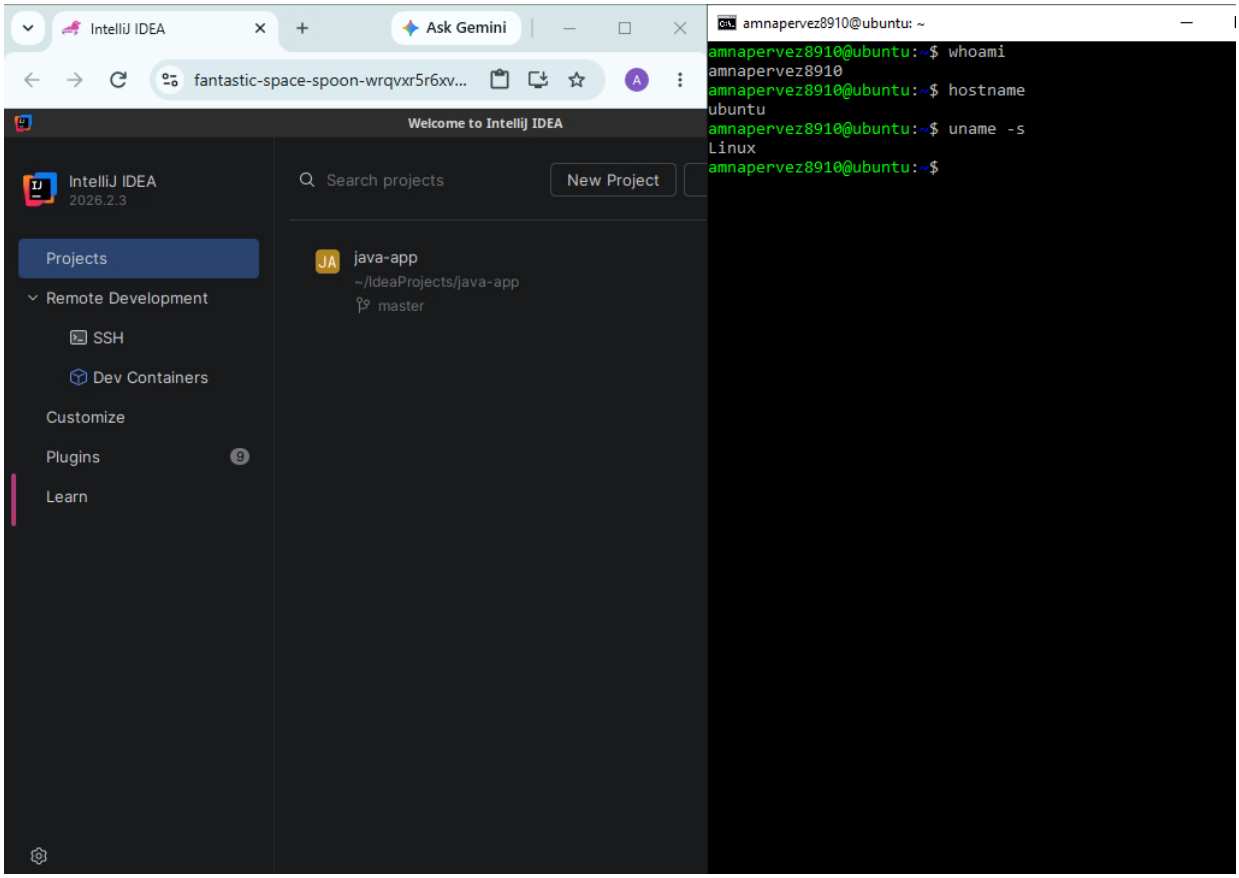
gradle_jar.png
build/libs has java-app-1.0-SNAPSHOT.jar.



gradle_running.png
Used JDK 16 and ran the jar with java -jar. The app started.

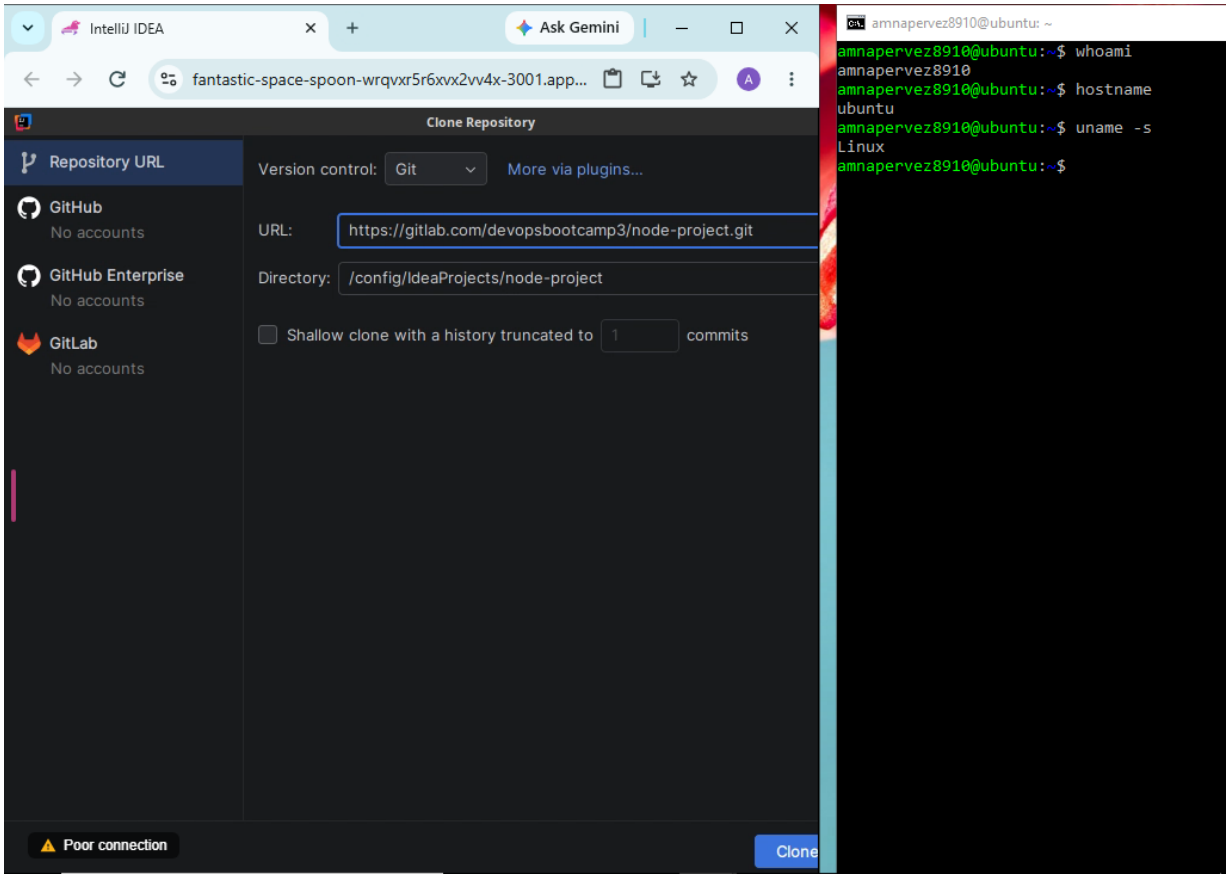


gradle_stopped.png
Stopped the app with Ctrl+C. The prompt returned.

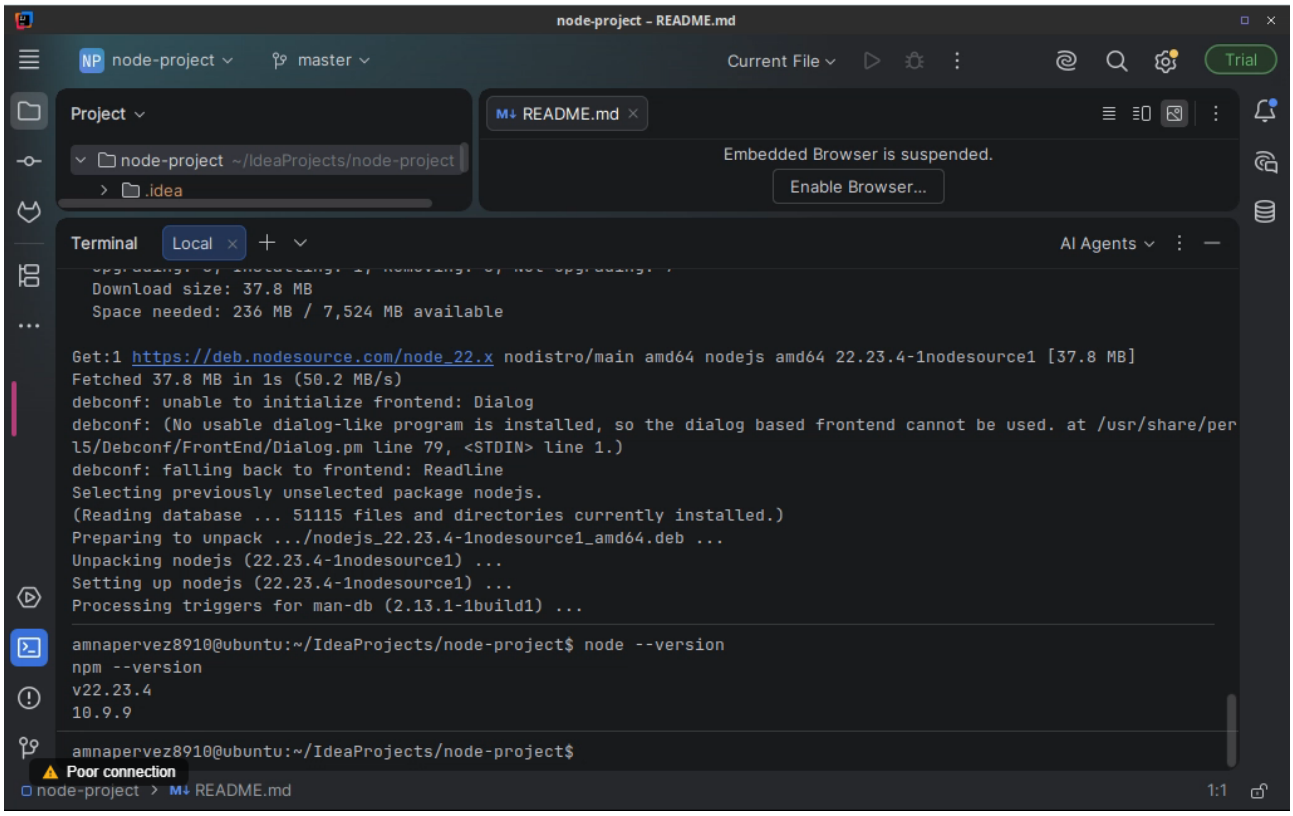


gradle_project_closed.png
Closed the Gradle project and returned to the Welcome screen.

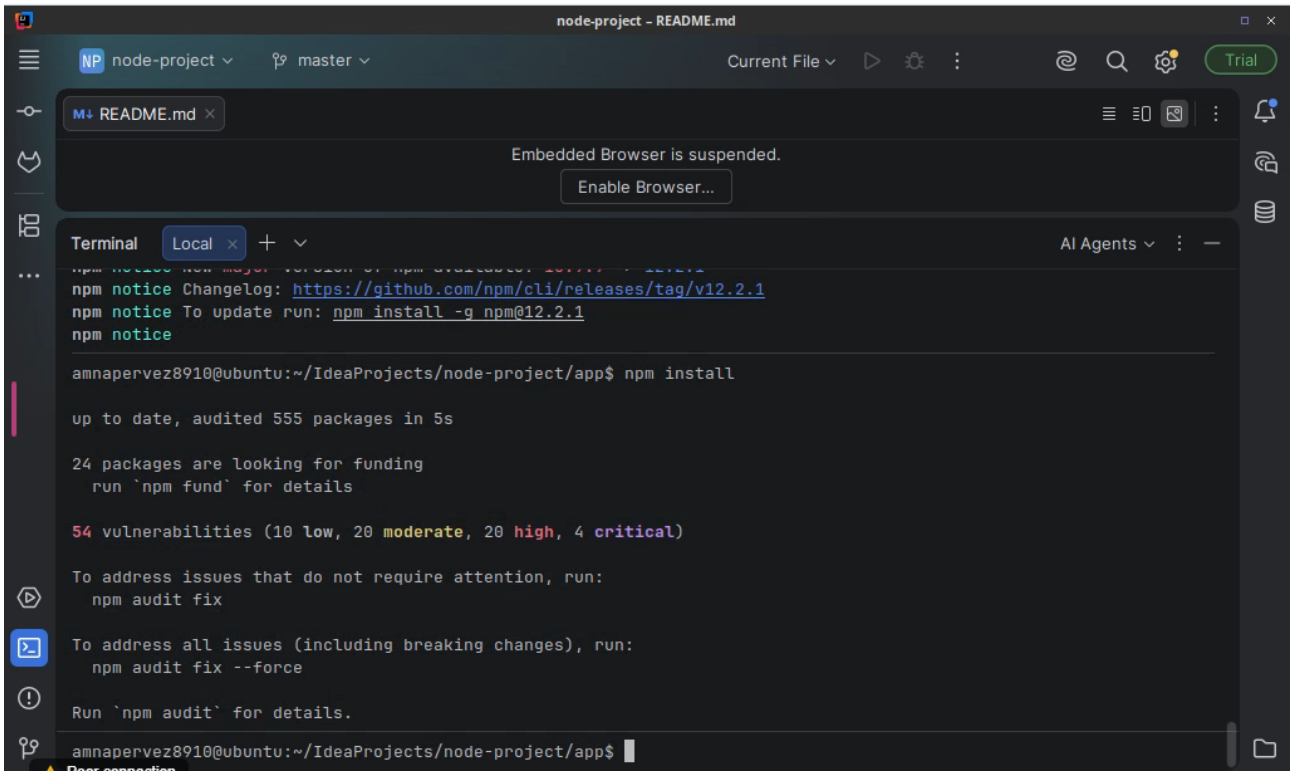
Task 4: Install and Run the Node.js Project



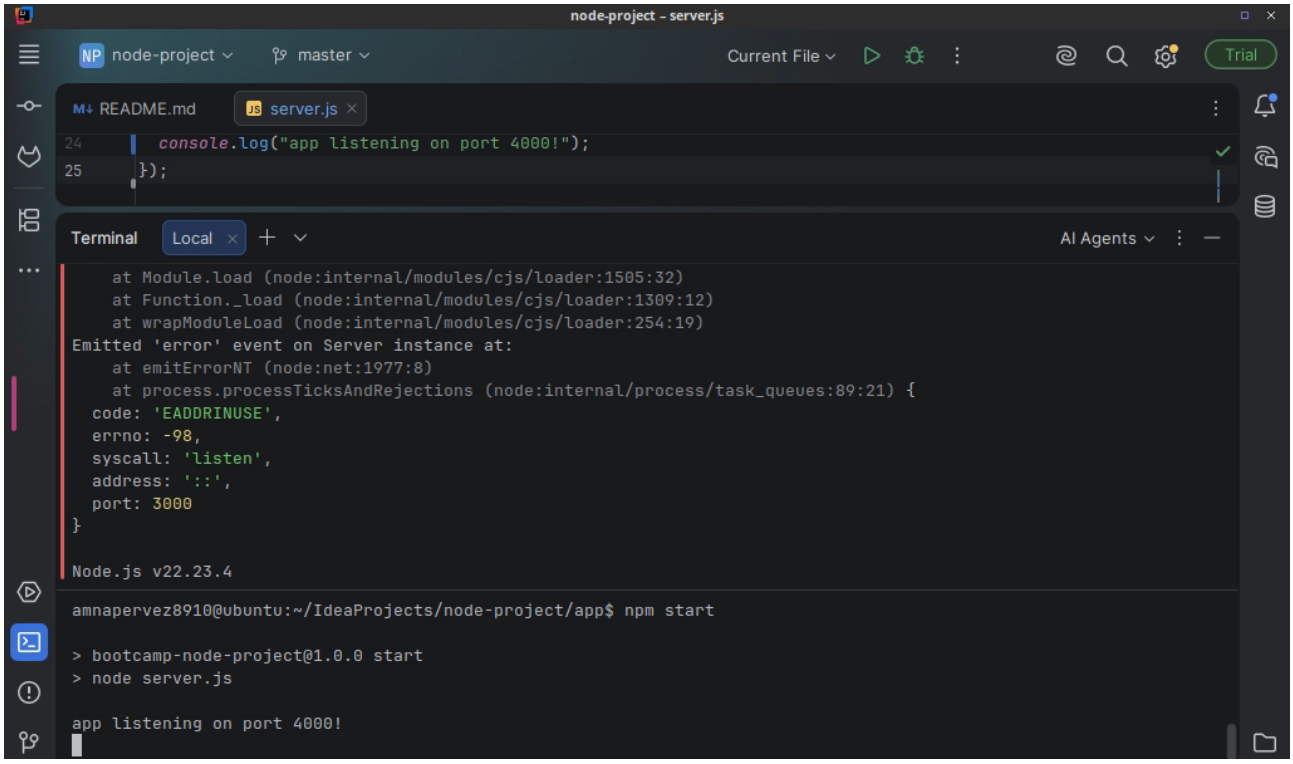
node_project_clone.png
Cloning the node-project from GitLab into /config/IdeaProjects/node-project.



nodejs_installed.png
Installed Node.js and npm. Versions: node v22.23.4, npm 10.9.9.

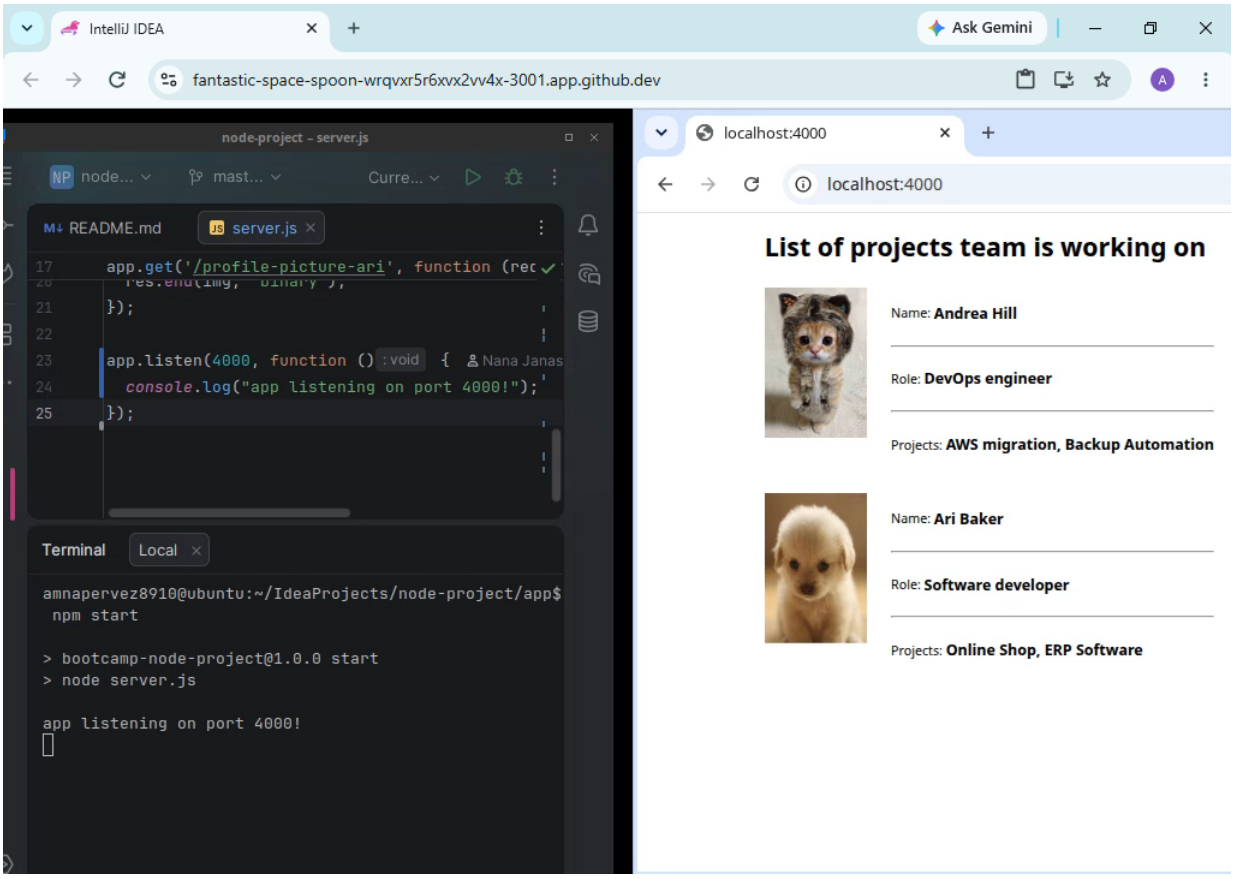


npm_install.png
Ran npm install inside the app folder. Dependencies installed.



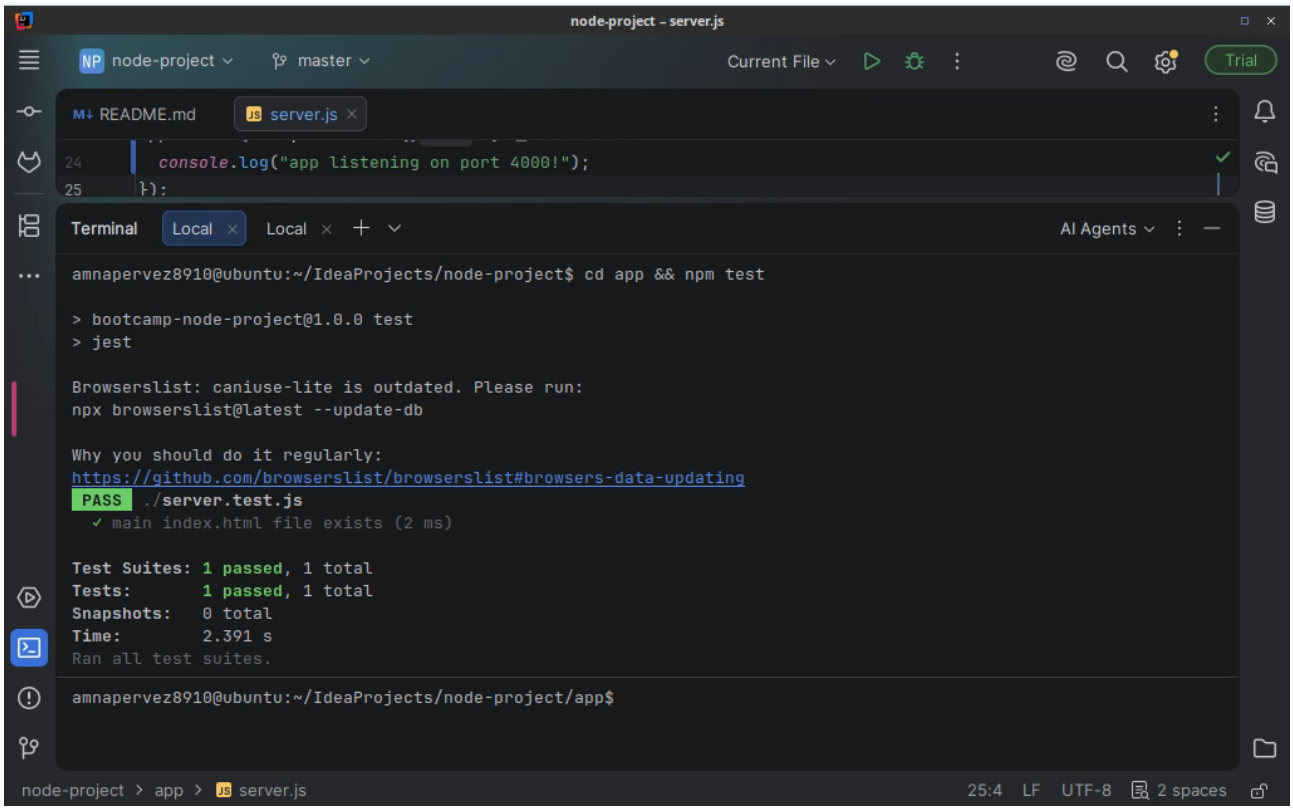
node_running.png

Port 3000 gave EADDRINUSE, so I changed server.js to port 4000. Now it says app listening on port 4000!

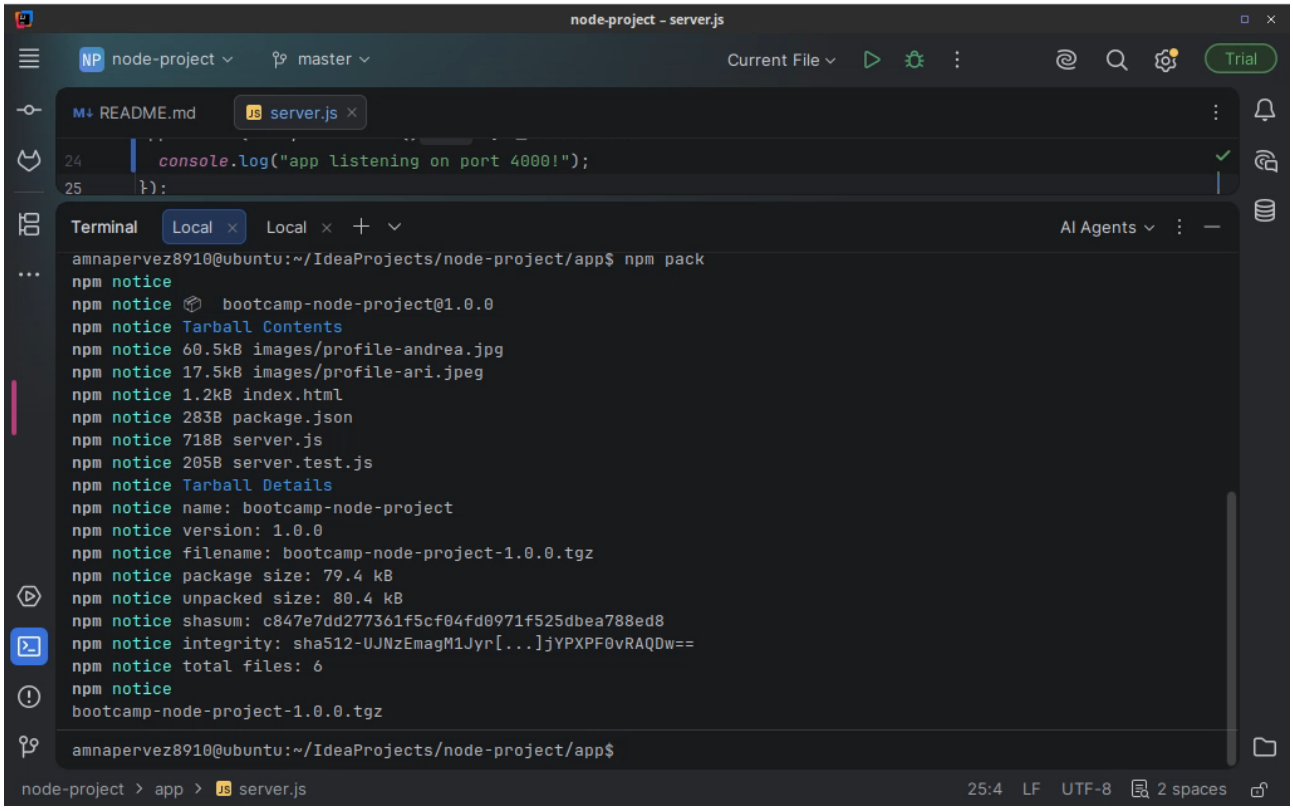


node_browser.png

Opened http://localhost:4000 in the browser inside the container. The project page and images are visible.

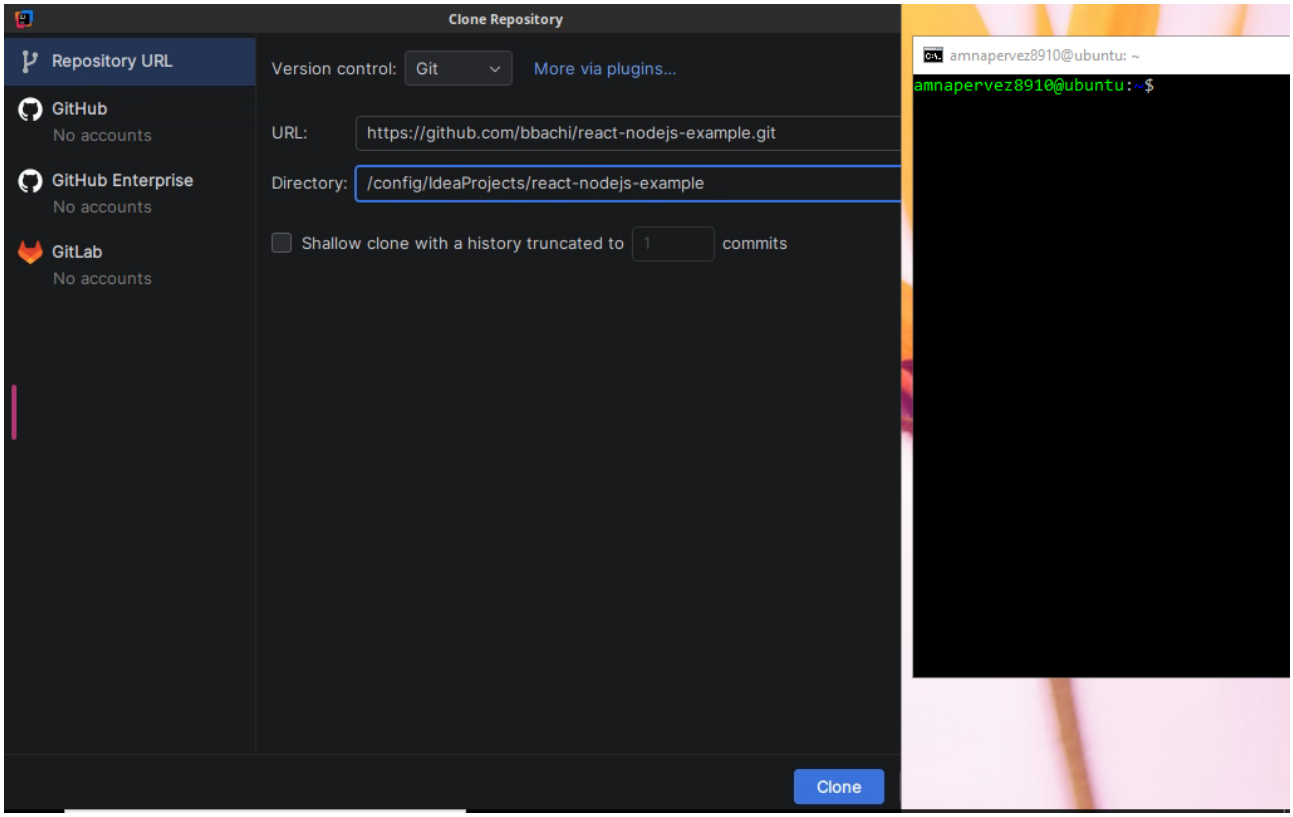


npm_test.png
Ran npm test. The Jest test passed.

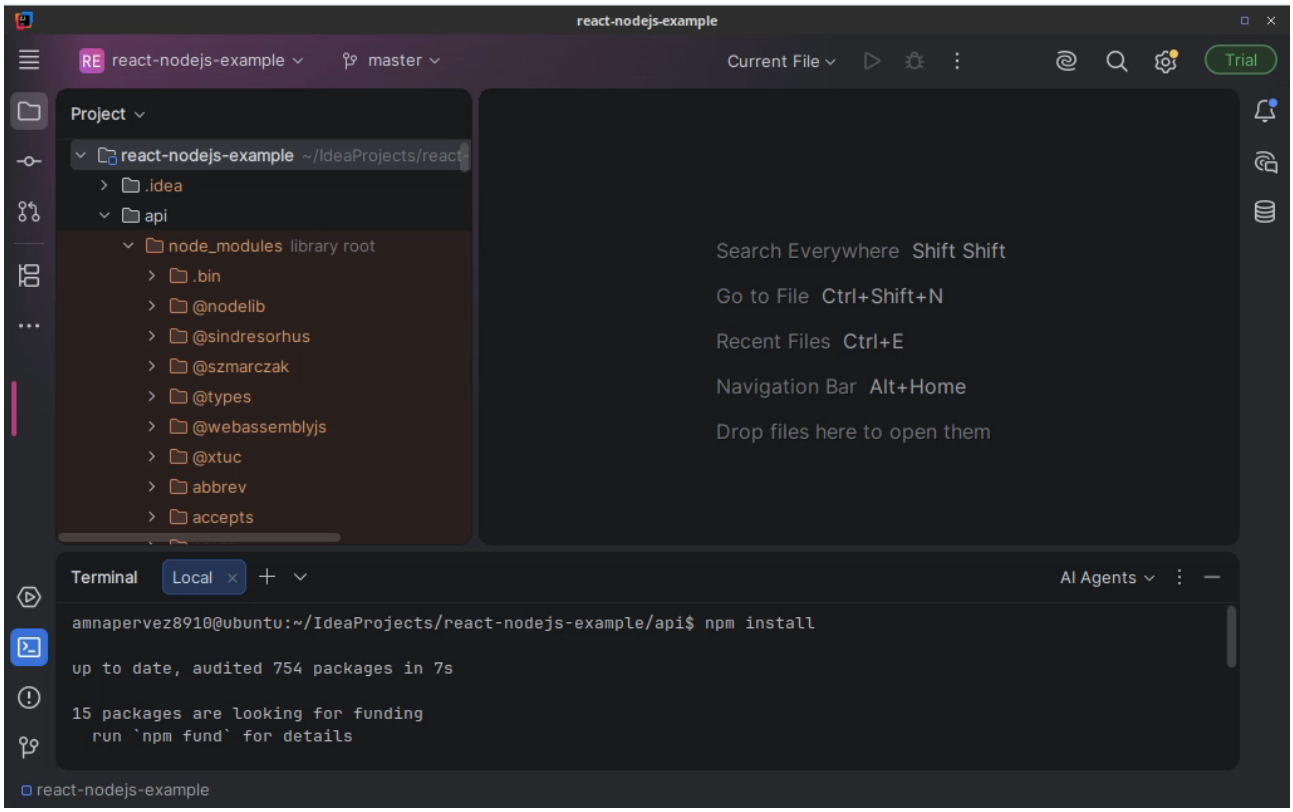


npm_pack.png
Ran npm pack. It created bootcamp-node-project-1.0.0.tgz.

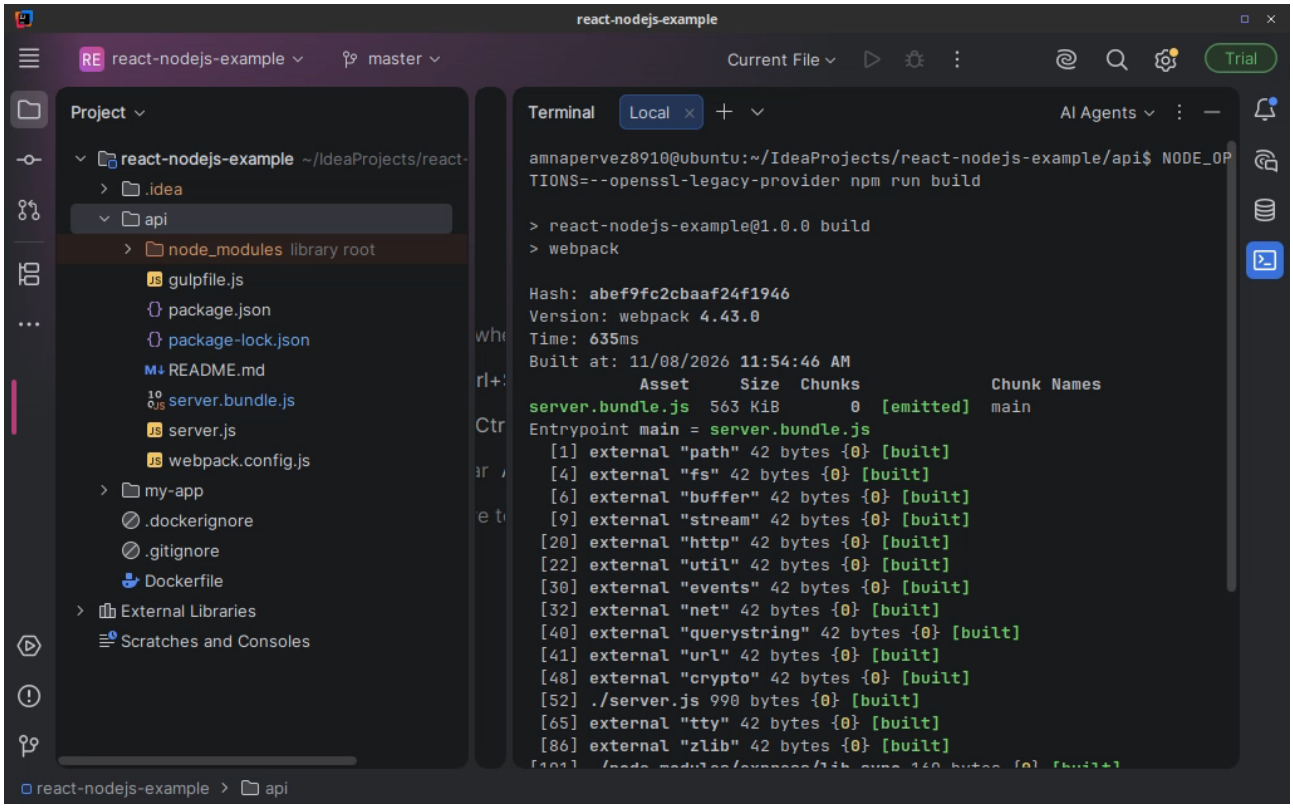
Task 5: Build the React and Node.js API Artifact



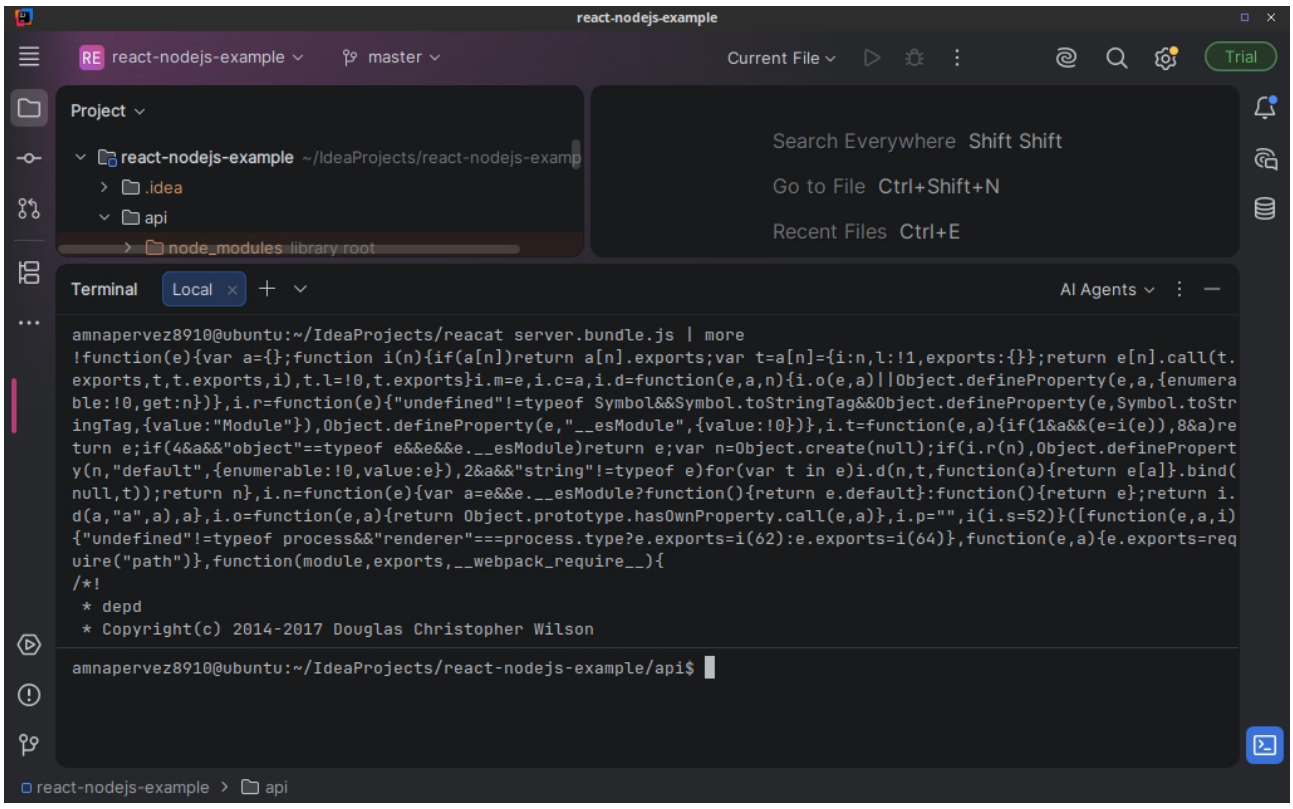
react_node_project_clone.png
Cloning react-nodejs-example from GitHub into /config/IdeaProjects/react-nodejs-example.



react_api_dependencies.png
Ran npm install in the api folder. node_modules is visible in the Project pane.

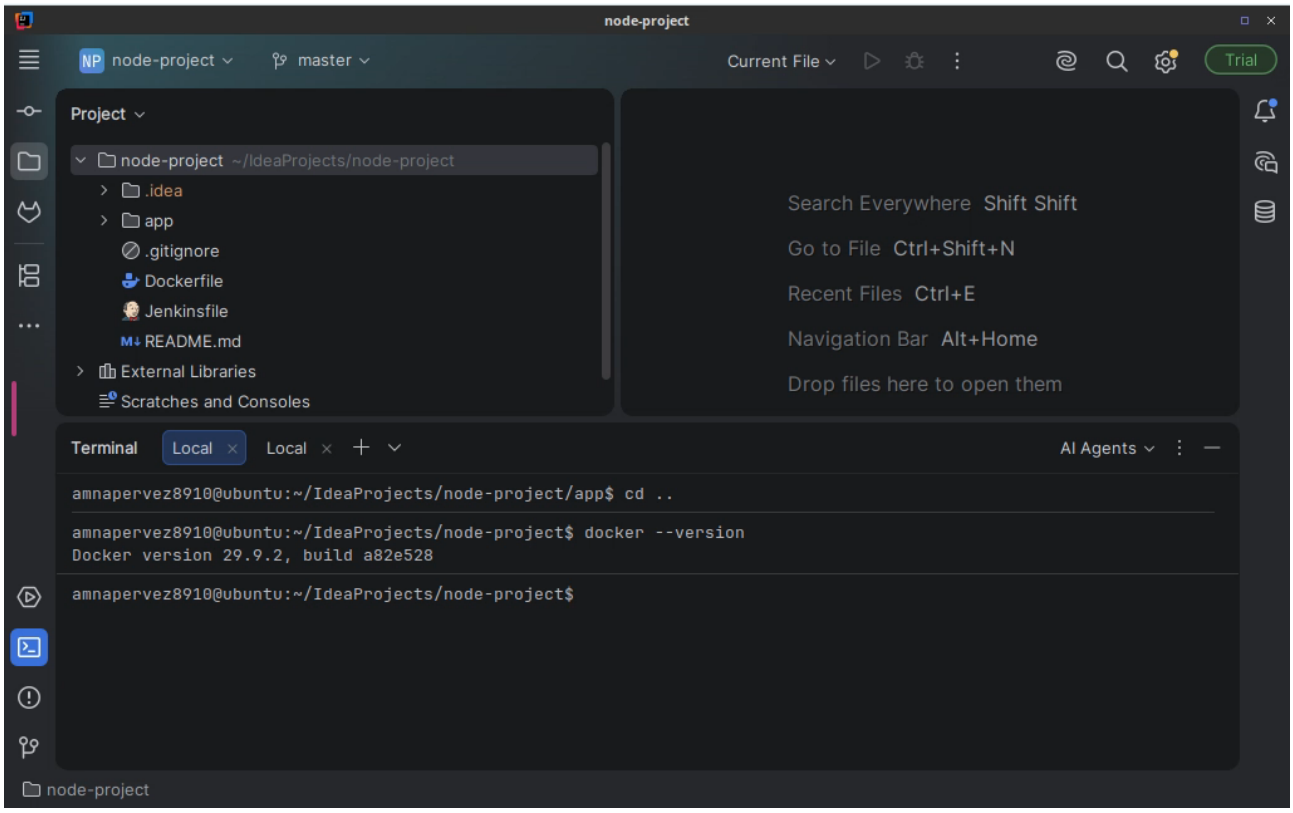


react_api_build.png
Ran npm run build (with the openssl legacy option). Webpack created server.bundle.js.

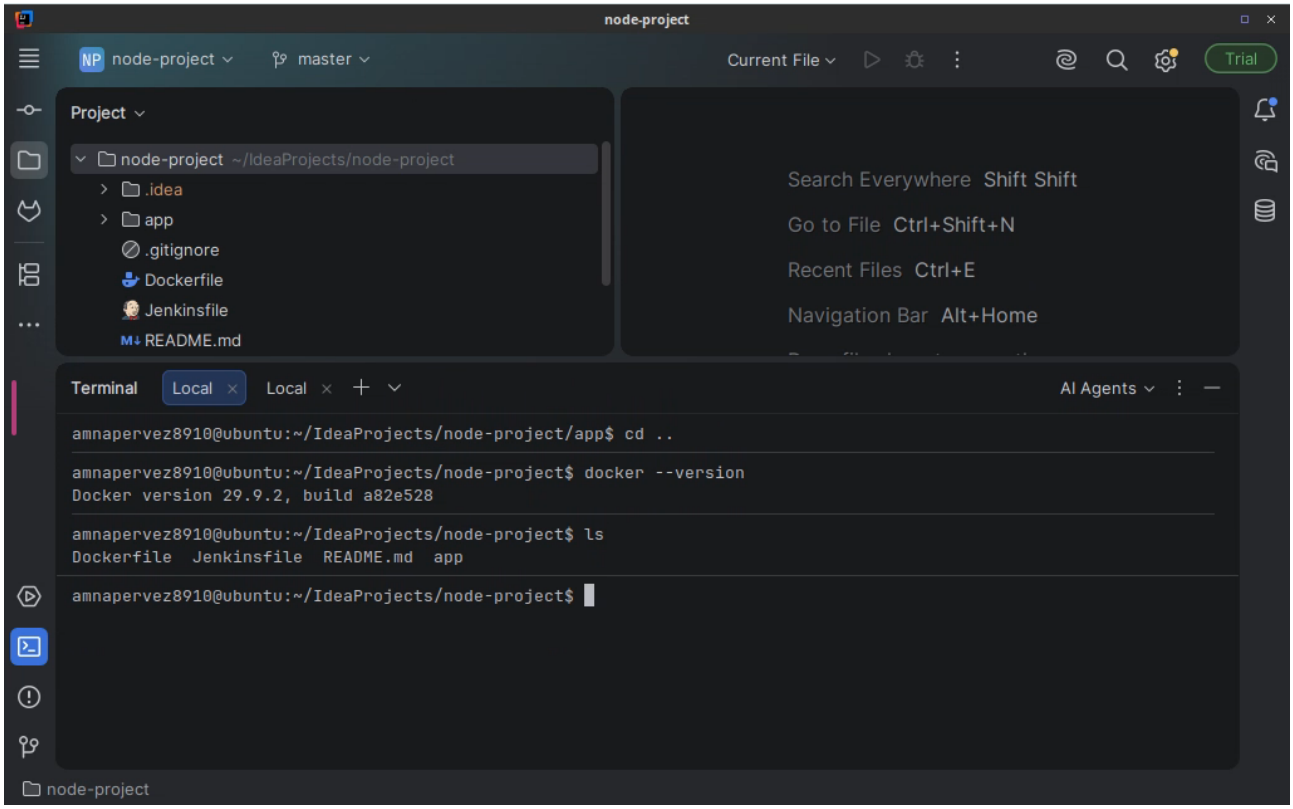


react_api_bundle.png
Viewed server.bundle.js using cat server.bundle.js | more.

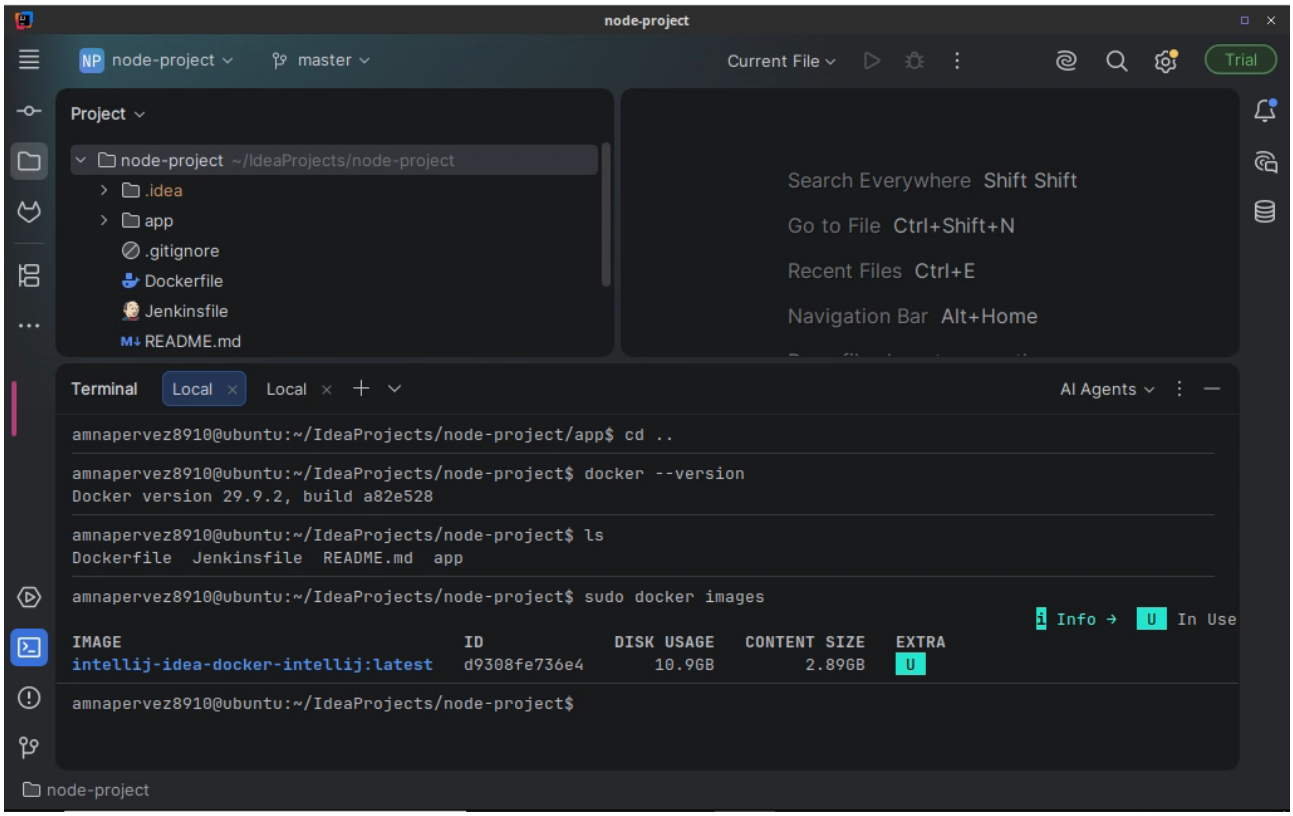
Task 6: Build a Docker Image for the Node.js Project



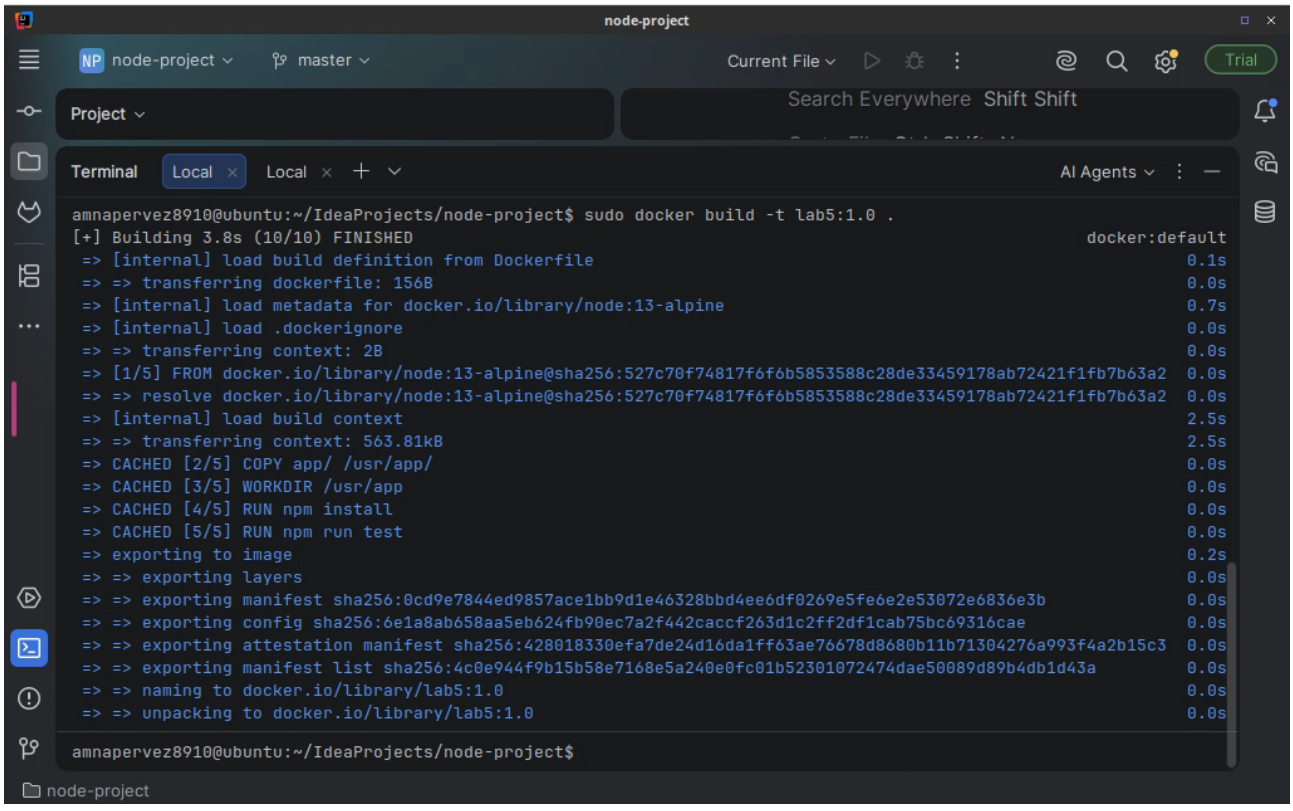
docker_installed.png
docker --version shows Docker version 29.9.2.



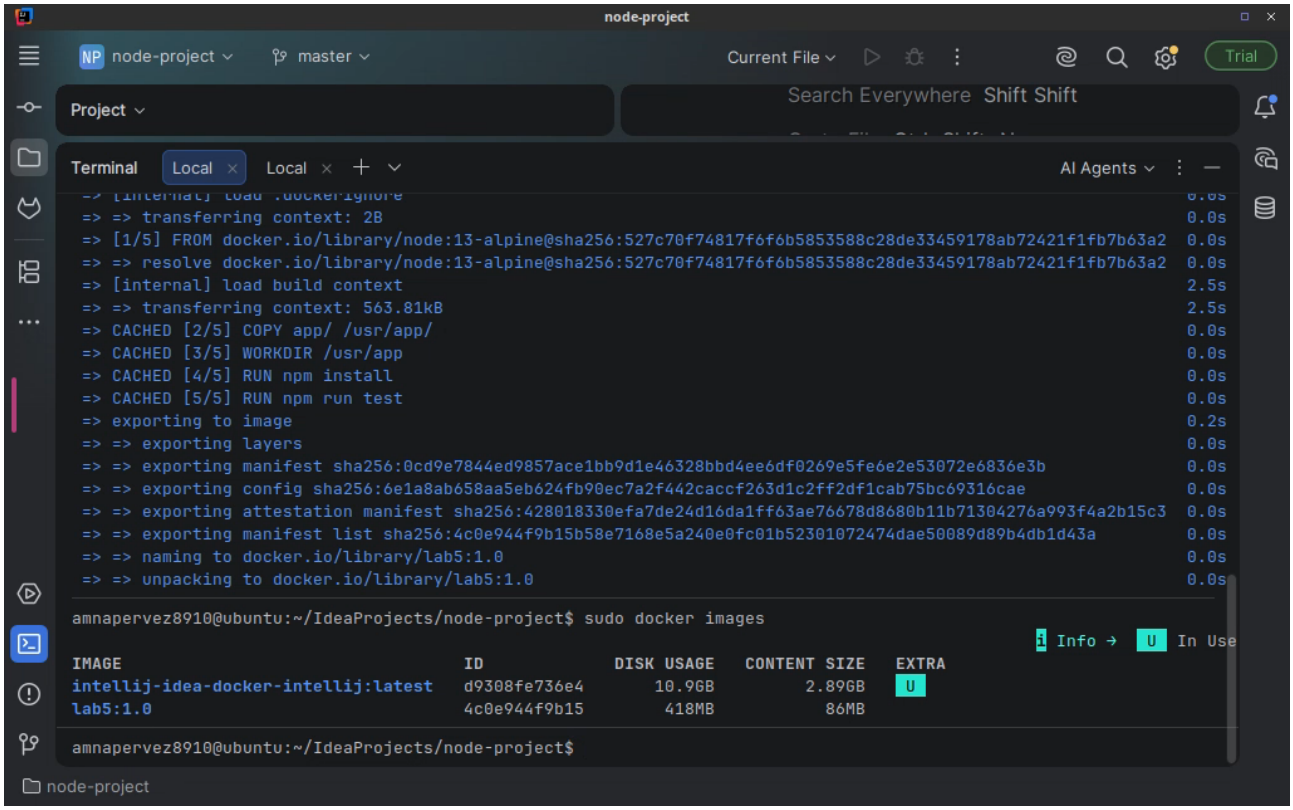
dockerfile_exists.png
ls shows the Dockerfile in the repository root.



docker_images_before.png
docker images before the build. lab5 is not there yet.



docker_build.png
Built the image with sudo docker build -t lab5:1.0 . and it finished successfully.



docker_images_after.png
docker images after the build. lab5 with tag 1.0 is now listed.

Summary

In this lab I ran IntelliJ IDEA in Docker, built and ran the Maven and Gradle projects with the Elasticsearch dependency, ran and packaged the Node.js project, built the React API bundle with webpack, and built the Docker image lab5:1.0.